7

Sound

In this chapter, we will learn about

- the properties of longitudinal waves
- the nature of sound
- audible and inaudible sound

OXFORD UNIVERSITY PRESS

7.1 Longitudinal waves

You learned in Chapter 4.1 that waves can be classified as transverse or longitudinal according to how the particles in the waves vibrate. In this part, you will learn more about the properties of longitudinal waves.

DSE goal

Describe wave motion in terms of amplitude, period, frequency, wavelength and wave speed.

Apply $f = 1/T$ and $v = f\lambda$ to solve problems.

DSE exam

13(1A)Q17, 15(1A)Q12, 20(1A)Q11, 21(1A)Q12

Video

→ This video demonstrates the Hands-on activity.

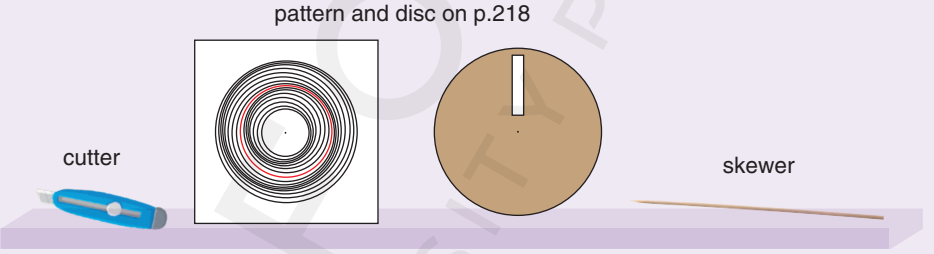
1 Describing a longitudinal wave

Longitudinal wave model

Hands-on activity

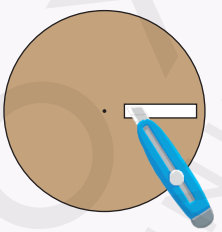
→ Worksheet is available on OUP web.

What you need



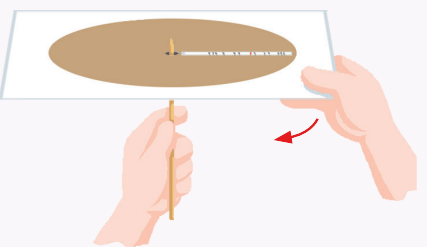
What to do

1



Cut out the square with the pattern as well as the disc. Cut a slit on the disc.

2



Place the disc on top of the square paper. Pin their centres together. Rotate the square paper while keeping the disc stationary. Observe the pattern through the slit.

What you find

How do the lines on the pattern move?

The wave model in the above activity demonstrates the motion of a longitudinal wave. It recalls that (Fig 7.1a on p.155):

1 A longitudinal wave consists of a series of **compressions** and **rarefactions**.

2 The particles on a longitudinal wave vibrate **along** the direction the wave travels.

→ This simulation shows the meaning of amplitude, wavelength, frequency and period for a longitudinal wave. The size of each quantity can be changed to show how the wave is affected.. It is available on OUP web in PC version (.exe) only.

Simulation

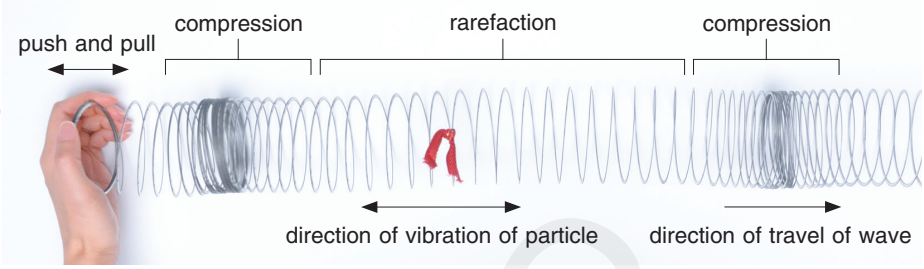


Fig 7.1a *A longitudinal wave.*

Cross-link

The terms are the same as those describing a transverse wave. See Ch 4.2.

Figure 7.1b shows the particles on a longitudinal wave travelling to the right at different times *t*. Table 7.1a shows the terms that describe a longitudinal wave.

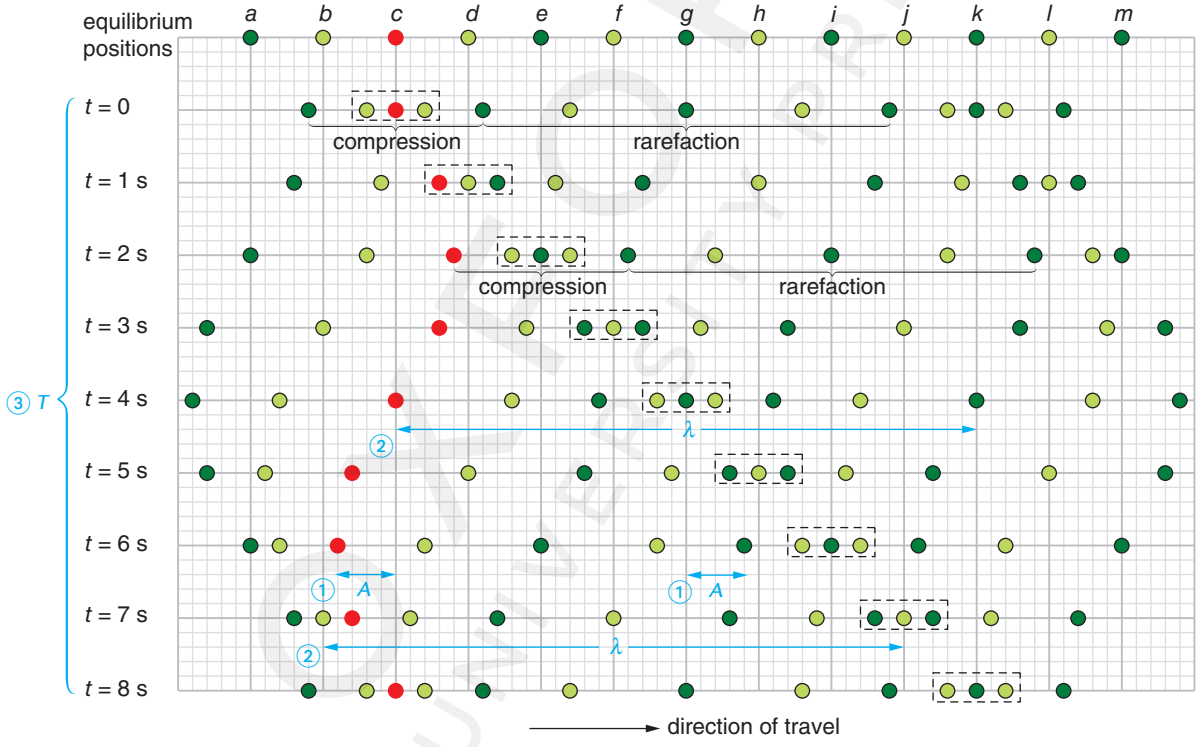


Fig 7.1b *Particles on a longitudinal wave at different times.*

Terms	Unit	Meaning
amplitude (A)	m	size of maximum displacement of a particle from its equilibrium position (see ① in Fig 7.1b)
wavelength (λ)	m	minimum distance over which the waveform repeats itself (see ② in Fig 7.1b)
period (T)	s	time for a particle to complete one vibration, or time for one complete cycle of a wave to be produced, or time for the wave to travel one wavelength (see ③ in Fig 7.1b)
frequency (f)	Hz	number of vibrations made by a particle in 1 s, or number of cycles of a wave produced in 1 s
wave speed (v)	m s^{-1}	distance travelled by the wave in 1 s

Table 7.1a *Terms describing a longitudinal wave.*

Particles are symmetric about the centres of compression and rarefaction.

You may find λ by measuring the distance between two successive centres of compression (or rarefaction).

Note that

1 All particles vibrate with the same period, frequency and amplitude.

2 Particles at the centres of compression (e.g. c and k in Fig 7.1c) or centres of rarefaction (e.g. g) are at their equilibrium positions.

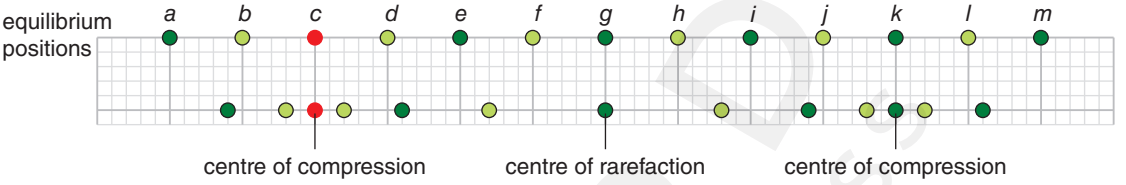
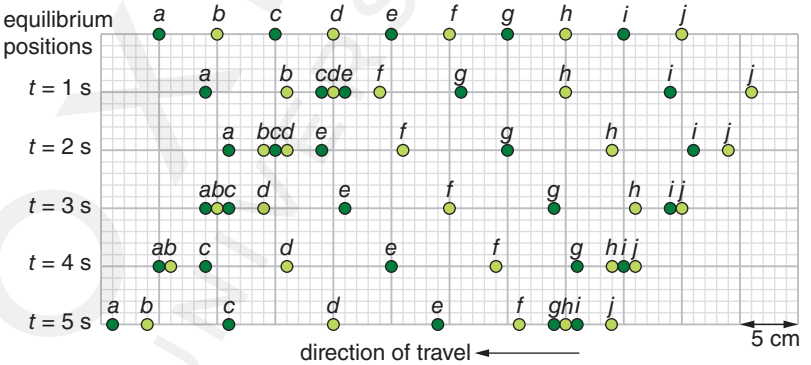


Fig 7.1c *Centres of compression and centres of rarefaction.*

3 Like a transverse wave, $f = \frac{1}{T}$ and $v = f\lambda$.

Example 1 | Describing a longitudinal wave

The figure below shows the particles on a longitudinal wave travelling to the left at different times t. Find the amplitude, wavelength, period, frequency and speed of the wave.



Solution

You may also look for a particle which is furthest away from the equilibrium position at a certain instant, e.g. particle b at t = 1 s.

Particle a is furthest away from its equilibrium position at t = 2 s.

$\therefore A = 6$ cm

At t = 3 s, b and j are two adjacent centres of compression.

$\therefore \lambda = 8 \times 5 = 40$ cm $= 0.4$ m

d completes half a vibration from t = 1 s to t = 5 s.

$\therefore T = 2 \times (5 - 1) = 8$ s

$f = \frac{1}{T} = \frac{1}{8} = 0.125$ Hz

Alternatively,
the wave travels 5 cm in 1 s.

$\therefore v = \frac{0.05}{1} = 0.05$ m s^{-1}

$v = f\lambda = (0.125)(0.4) = 0.05$ m s^{-1}

Quick check 1

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 The figure shows the positions of the particles on a longitudinal wave at different times t. The wave travels to the right. Find the wave's

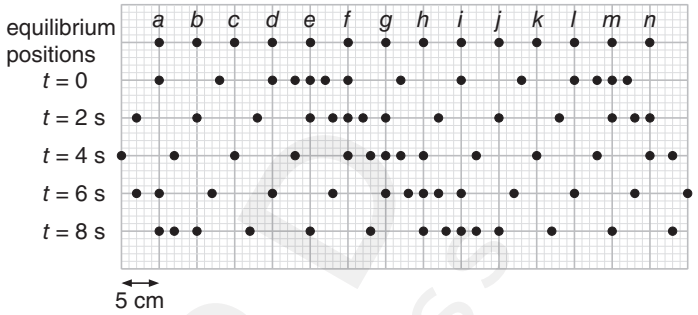
(a) amplitude,

(b) speed,

(c) wavelength,

(d) period,

(e) frequency.



Example 2 Possible frequencies of a wave

A longitudinal wave travels to the right along a slinky spring. Three snapshots of the spring taken at 0.6 s intervals are as shown (Fig a). Suggest the three minimum possible frequencies of the wave. Explain your answer.

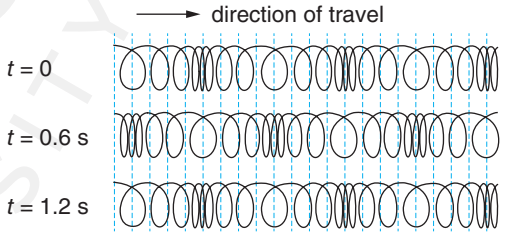


Fig a

Solution

From $t = 0$ to $t = 0.6$ s, the wave may have travelled 0.5λ, 1.5λ (Fig b), 2.5λ, etc.

If the wave has travelled 0.5λ in 0.6 s,

$$v_1 = \frac{0.5\lambda}{0.6} = 0.833\lambda$$

If the wave has travelled 1.5λ in 0.6 s,

$$v_2 = \frac{1.5\lambda}{0.6} = 2.5\lambda$$

If the wave has travelled 2.5λ in 0.6 s,

$$v_3 = \frac{2.5\lambda}{0.6} = 4.17\lambda$$

By $f = \frac{v}{\lambda}$,

$f_1 = 0.833$ Hz

$f_2 = 2.5$ Hz

$f_3 = 4.17$ Hz

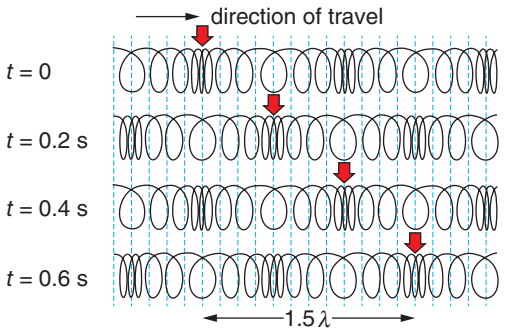


Fig b

Revision 7 Q6 (p.190)

DSE goal

Describe wave motion in terms of phase and displacement.

DSE exam

12(1A)Q15, 17(1A)Q14

2 Motion of particles

The motion of the particles on a longitudinal wave is like that on a transverse wave.

1 Particles at the **equilibrium positions** (centres of compression or rarefaction) are moving at the **highest speed**.

They are about to reverse their moving directions.

2 Particles at their **maximum displacements** are **momentarily at rest**.

Centres of compression are $n\lambda$ apart. A centre of compression and a centre of rarefaction are $(n + 0.5)\lambda$ apart.

3 Two particles separated by $n\lambda$ (where n is an integer) vibrate in phase. Two particles whose **equilibrium positions** are separated by $(n + 0.5)\lambda$ vibrate in antiphase.

If the wave travels to the left, a particle has the same displacement as the particle on its right in the previous instant.

4 Suppose a wave travels to the right. A particle has the same displacement as the particle on its left in the previous instant (Fig 7.1d).

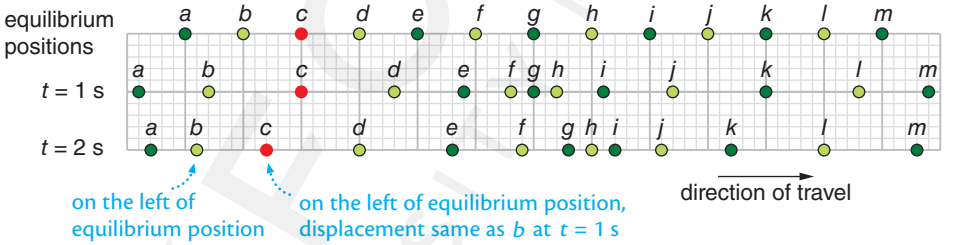
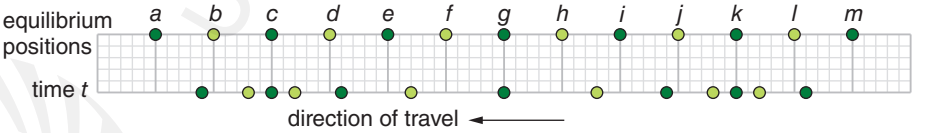


Fig 7.1d *Direction of motion of particles.*

Example 3 Motion of particles

The positions of the particles on a longitudinal wave at time t are as shown. The wave is travelling to the left.



(a) Name a particle vibrating in phase with k.

(b) Name a particle vibrating in antiphase with h.

(c) Name a particle that is momentarily at rest at time t.

Solution

(a) c

(b) d or l

(c) a, e, i or m

Tip

The equilibrium positions of g and c are 0.5λ apart, as is the case with the equilibrium positions of h and d.

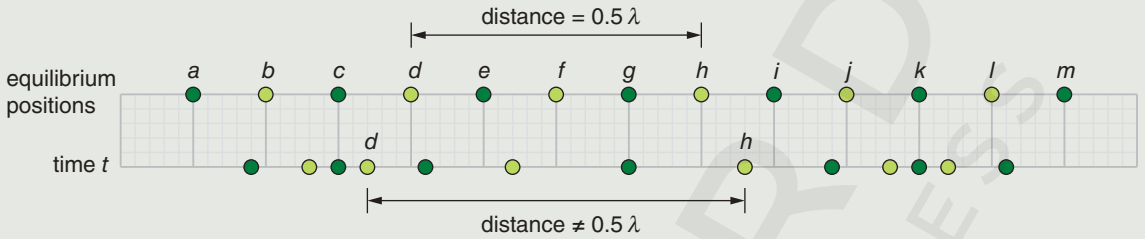
Caution

c, g and k are moving at the highest speed at time t. They are not momentarily at rest.

Know more

Distances and phases of particles

While a longitudinal wave is travelling, particles vibrating in phase are always at a distance $n\lambda$ from each other. However, particles vibrating in antiphase are not necessarily at a distance $\left(n + \frac{1}{2}\right)\lambda$ from each other due to their 'horizontal displacement'. Below is an example.



Google Forms Microsoft Forms

Quick check 2

→ E-version in OUP Exercise Platform is also available.

1 Determine whether the following statements about a longitudinal wave are true.

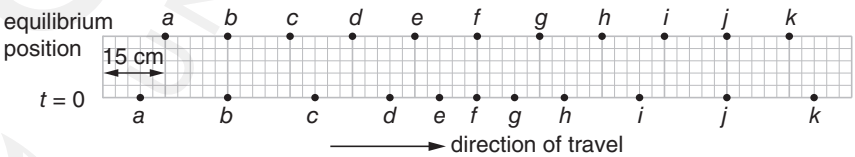
(a) Particles separated by 2λ always vibrate in phase. (T/F)

(b) When two particles vibrate in antiphase, the distance between them is constant. (T/F)

(c) A particle at a centre of rarefaction is moving at the highest speed. (T/F)

(d) A particle is momentarily at rest when it is furthest away from other particles. (T/F)

2 In the figure below, the positions of particles a to k on a longitudinal wave at time $t = 0$ are as indicated. The wave travels to the right.



(a) Which particle is momentarily at rest at $t = 0$?

A d B e C f

(b) Which particle is moving at the highest speed?

A d B e C f

(c) Which particle vibrates in phase with b?

A f B h C j

(d) Which particle vibrates in antiphase with e?

A g B i C k

(e) Which particle is moving to the right at $t = 0$?

A c B d C e

DSE goal

Present information on *s*–*t* and *s*–*d* graphs for travelling waves.

DSE exam

19(1A)Q14, 23(1B)Q5

3 Graphical description

Like transverse waves, displacement–time graphs (*s*–*t* graphs) and displacement–distance graphs (*s*–*d* graphs) can be used to describe longitudinal waves.

a Displacement–time graph

All the particles on a travelling longitudinal wave undergo vibration. The displacement *s* of a particle from its equilibrium position changes continuously with time *t* (Fig 7.1e). Let us take particle *c* as an example, with the displacement to the right taken as positive (Table 7.1b).

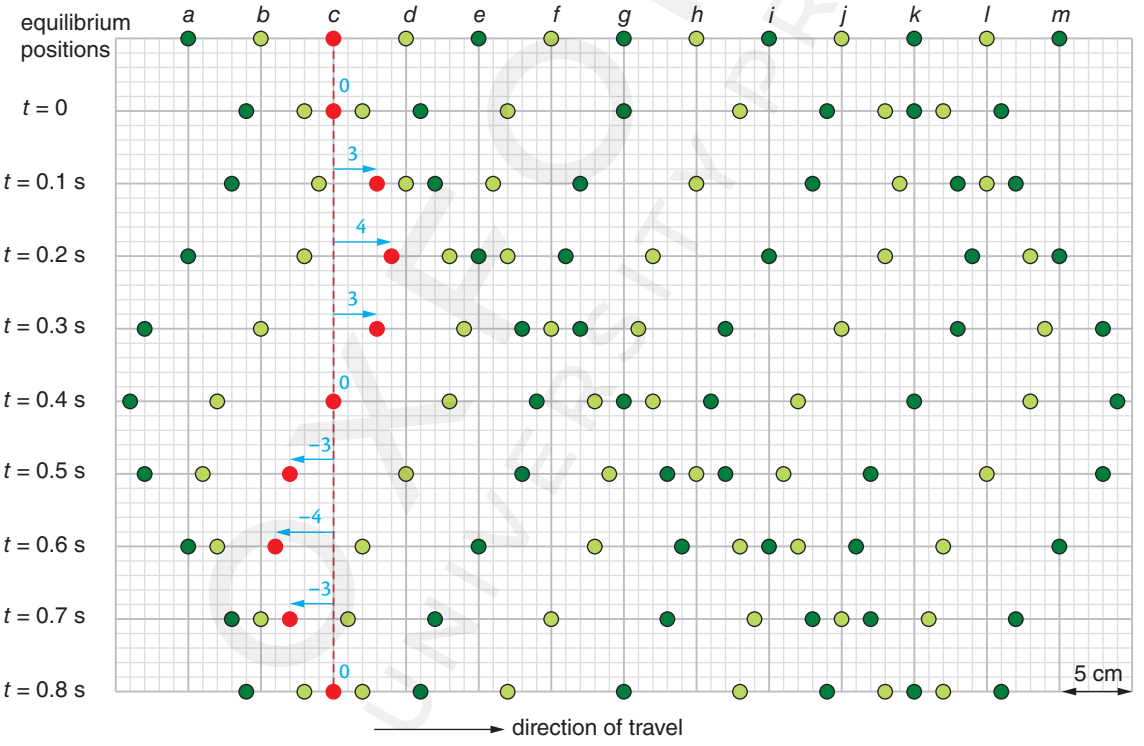


Fig 7.1e *A longitudinal wave at different times t.*

t / s	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8
s / cm	0	3	4	3	0	–3	–4	–3	0

Table 7.1b *The displacements of c.*

By plotting *s* against *t*, an *s*–*t* graph of particle *c* is produced (Fig 7.1f on p.161). From this graph, we can read the amplitude *A* and the period *T* of the particle vibration and the wave.

Note that an *s*–*t* graph shows the displacements of **a particular particle** at **different instants**.

At t = 0.2 s and 0.6 s, c is at its maximum displacement and is momentarily at rest (slope of graph = 0). At t = 0, 0.4 s and 0.8 s, c is at its equilibrium position and its speed is the highest (slope of graph has largest magnitude).

Ts may ask Ss to draw the s–t graphs of other particles in Fig 7.1e.

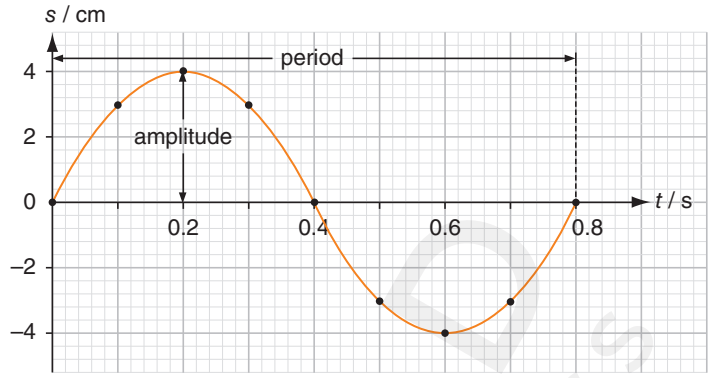
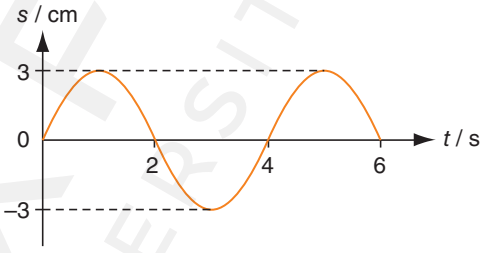


Fig 7.1f *The s–t graph of particle c.*

Example 4 The *s*–*t* graph of a longitudinal wave

The figure below shows the s–t graph of particle P on a longitudinal wave travelling to the right. The displacement towards the right is taken as positive.



(a) Find the amplitude of the wave.

(b) Find the frequency of the wave.

(c) Describe the motion of P at

(i) t = 0,

(ii) t = 1 s.

Solution

(a) 3 cm

(b) $f = \frac{1}{T} = \frac{1}{4}$ = 0.25 Hz

(c) (i) It is moving to the right.

(ii) It is momentarily at rest.

Tip

s changes from 0 to a positive value just after t = 0, so P is moving in the positive direction. Pay attention to the positive direction defined in the question. Do **not** answer upwards.

Think about...

How will the answers to (c) change if the wave travels towards the left and the displacement towards the left is taken as positive?

Google Forms Microsoft Forms

Quick check 3

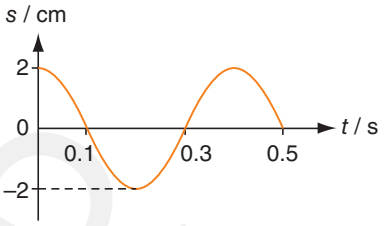
→ E-version in OUP Exercise Platform is also available.

1 The *s*–*t* graph of particle *P* on a longitudinal wave is as shown. The displacement towards the left is taken as positive.

(a) What is the amplitude of the wave?

(b) What is the frequency of the wave?

(c) Describe the motion of *P* at (i) $t = 0$ and (ii) $t = 0.1$ s.



b Displacement–distance graph

At a certain time *t*, the displacements of the particles at different positions on a longitudinal wave are different (Fig 7.1g). To indicate the positions of the particles, the distances between their **equilibrium positions** and a reference point are measured.

Let us take the equilibrium position of particle *A* as the reference point, and the displacement towards the right as positive. Table 7.1c records the displacements *s* of the particles at different distances *d*.

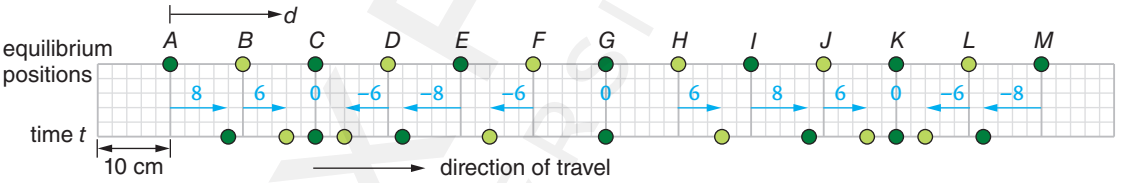


Fig 7.1g *A longitudinal wave at time t.*

Particles	*A*	*B*	*C*	*D*	*E*	*F*	*G*	*H*	*I*	*J*	*K*	*L*	*M*
***d* / cm**	0	10	20	30	40	50	60	70	80	90	100	110	120
***s* / cm**	8	6	0	–6	–8	–6	0	6	8	6	0	–6	–8

Table 7.1c *The displacements s of different particles.*

By plotting *s* against *d*, the *s*–*d* graph at this instant is produced (Fig 7.1h). We can read the amplitude *A* and the wavelength λ from the graph.

Note that an *s*–*d* graph shows the displacements of **different particles** at **a certain instant**.

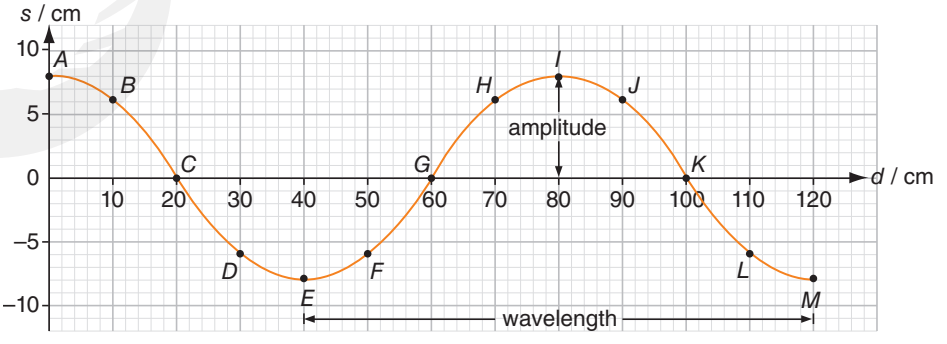


Fig 7.1h *The s–d graph of a longitudinal wave.*

It is a convention to take rightward direction to be positive when it is not specified. If the leftward direction is taken to be positive, the *s*–*d* graph obtained would be inverted compared to Figure 7.1h.

→ Teaching notes and exercises are available on OUP web.

Skill Graph

Emphasize that the rightward direction is taken to be positive in the discussion below.

Finding the position of the centre of compression or rarefaction from an *s*–*d* graph

The intercepts on the distance axis give the positions of the centres of compression or rarefaction. In Figure 7.1h, if the curve goes from positive to negative through the axis, the intercept corresponds to a centre of compression (Fig a). If the curve goes from negative to positive, the intercept corresponds to a centre of rarefaction (Fig b).

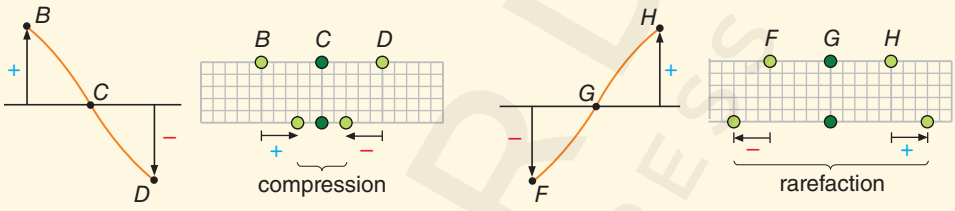
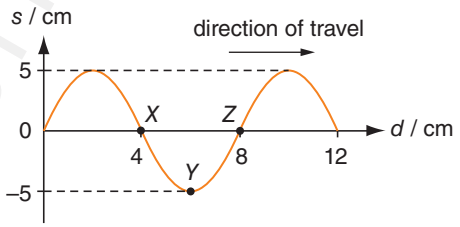


Fig a *Fig b*

Example 5 The *s*–*d* graph of a longitudinal wave

A longitudinal wave travels to the right. The *s*–*d* graph of the wave at time $t = 0$ is as shown. The displacement to the right is taken as positive. *X*, *Y* and *Z* are particles on the wave.



(a) Find the wavelength of the wave.

(b) Name a particle at the centre of compression at $t = 0$.

(c) Describe the motion of *Z* at $t = 0$.

Solution

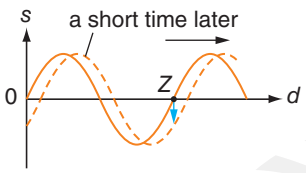
(a) 8 cm

(b) *X*

(c) *Z* is moving to the left.

Tip

The *s*–*d* graph after a very short time:



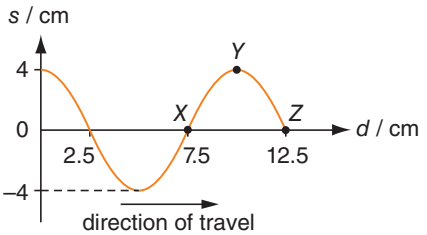
∴ *Z* is moving in the negative direction, i.e. towards left. ▶

Quick check 4

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 A longitudinal wave travels to the right. Its *s*–*d* graph at a certain instant is as shown, with the displacement to the right taken as positive. *X*, *Y* and *Z* are particles on the wave.



(a) What is the wavelength of the wave?

(b) Is *X* moving to the left or to the right?

(c) Which particle is at the centre of rarefaction?

Exercise 7.1

(For Q1–2.) Consider some particles on a longitudinal wave. Their equilibrium positions are shown in Figure a. Figure b shows their positions at time t. The wave is travelling to the right.

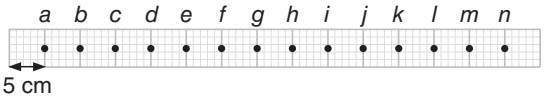


Fig a

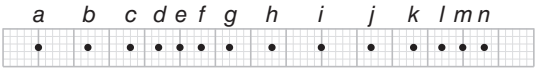


Fig b

★**1** Which of the following statements are correct?

(1) Its wavelength is 20 cm.

(2) Particle c vibrates in phase with particle k.

(3) All particles would have travelled the same distance in half a period.

A (1) and (2) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

★**2** Which of the following statements about the particles at time t are correct?

(1) Particle c is momentarily at rest.

(2) Particle e is moving at the highest speed.

(3) Particle d is moving towards the right.

A (1) and (2) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

★**3** A travelling longitudinal wave is generated along a slinky spring at a frequency of 4 Hz. The figure below shows the wave at a certain time (Fig c).

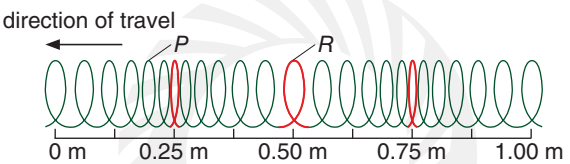


Fig c

(a) Estimate the wavelength of the wave.

(b) Find the wave speed.

(c) At the instant shown, in which direction does coil R move?

(d) Find the displacement of R from its equilibrium position after 0.5 s.

(e) Coils P and Q vibrate in antiphase. Label all possible positions of Q in Figure c.

★**4** A longitudinal wave is travelling towards the right along a long spring. The positions of some equally spaced particles on the spring at different instants are shown in Figure d.

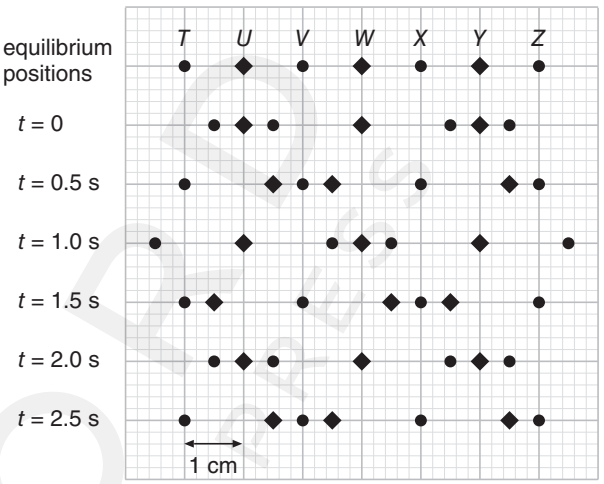


Fig d

(a) Find the wave speed.

(b) Sketch the s–t graphs for particles T and V from $t = 0$ to $t = 2.0$ s.

(c) Sketch the s–d graphs for the wave at $t = 0.5$ s and 1.0 s.

★**5** The s–d graph of a longitudinal wave at time $t = 0$ is as shown (Fig e). The wave travels to the right and the displacement towards the right is taken as positive. X, Y and Z are three particles on the wave.

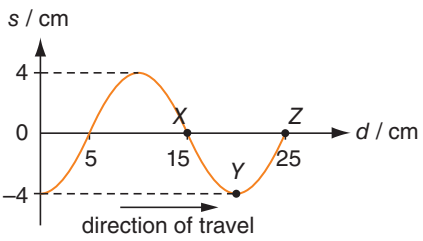


Fig e

(a) Describe the motion of X, Y and Z at this instant.

(b) Name a particle at the centre of rarefaction at this instant.

(c) Y reaches the equilibrium position for the first time at $t = 0.1$ s.

(i) Find the frequency of the wave.

(ii) Find the speed of the wave.

(iii) Sketch the s–t graph for X from $t = 0$ to $t = 0.5$ s.

7.2 Nature of sound

1 Sound as a wave

DSE goal
Realize sound as example of longitudinal wave.

When a candle goes out, the smoke rises and follows a smooth path (Fig 7.2a). However, when a loudspeaker is put in front of the candle and produces sound, the smoke rises in a series of zigzags (Fig 7.2b). Why?

Fig 7.2a *Smoke rises in a smooth path.*

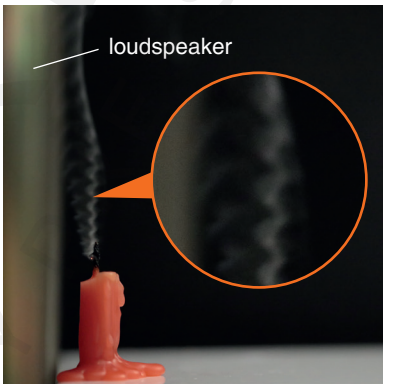


Fig 7.2b *Smoke rises in a series of zigzags.*

Make sure there is no wind when doing this demonstration. Air conditioners and fans should be switched off.

You learned the answer in Chapter 4.1: sound is a longitudinal wave. When the loudspeaker is turned on, its cone vibrates. Air particles in front of the cone are forced to vibrate to and fro. This produces a sound wave, which carries energy and travels outwards (Fig 7.2c). We hear the sound when the sound wave enters our ears.

In Figure 7.2b, the smoke particles vibrate in the horizontal direction when rising. This results in the zigzags.

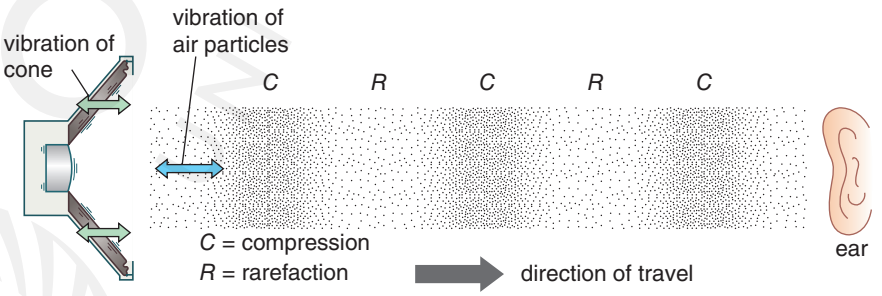


Fig 7.2c *Sound waves produced by a loudspeaker.*

The motion of the air particles can also be shown by putting a candle flame in front of the loudspeaker.

Video

→ This video shows a flickering candle flame placed in front of a sound-emitting loudspeaker.

Know more

Sound and vibration

All sound sources produce sound by vibrations. You can feel the vibration of your vocal cords when you speak.

DSE goal

Realize the need for a medium for sound transmission.

DSE exam

14(1A)Q19, 17(1A)Q15, 20(1B)Q6

2 Transmission of sound

Sound is a **mechanical wave**. It needs a medium for transmission. In addition to gas, sound can also travel through solids and liquids (Fig 7.2d). Particles in the medium vibrate as sound travel through them.

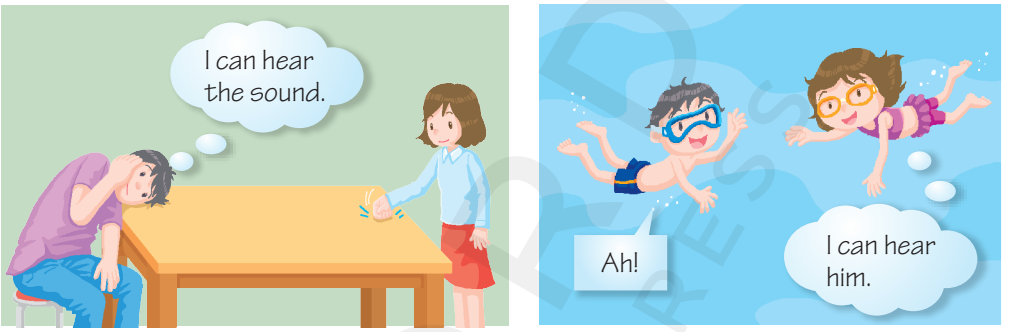


Fig 7.2d *Sound can travel through solids and liquids.*

However, sound waves cannot travel through a vacuum. We can demonstrate this idea by using the set-up in Figure 7.2e. When air inside the glass jar is pumped out, less and less sound can be heard from the electric bell.

Astronauts in space cannot talk to each other directly even if they are very close to each other. A radio system making use of electromagnetic waves is needed for them to communicate (Fig 7.2f).

▶ EM waves do not require a medium for transmission.

✎ The sound of the electric bell never disappears completely because some of it can still be transmitted through the supporting string of the bell to the outside.

✎ In movies, a loud sound is often produced when a spacecraft is travelling in space. Ts may ask Ss if this is correct.

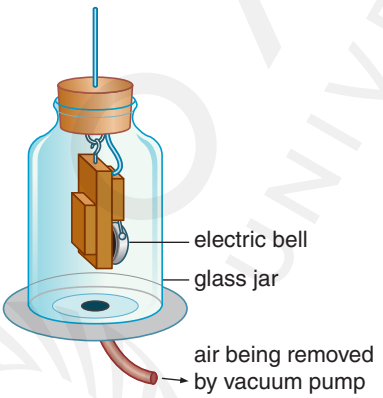


Fig 7.2e *The sound heard becomes weaker and weaker as air is pumped out.*

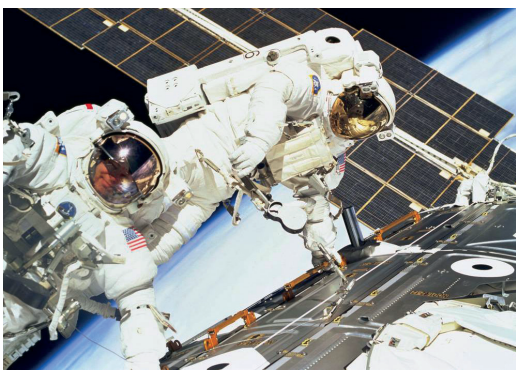


Fig 7.2f *Astronauts have to talk to each other via a radio system in space.*

The speed of sound depends on the medium it travels through. In general, sound travels faster in solids than in liquids, and in turn, faster in liquids than in gases.

✎ This statement is only generally but not absolutely true. The speed of sound depends on many factors, including the elastic properties and the density of the medium, the frequency of the sound etc., which are outside the scope of the syllabus.

Speed of sound in different media:

$$v_{solid} > v_{liquid} > v_{gas}$$

Note that the speed of sound in air is **independent** of the **frequency**.

Table 7.2a shows the speed of sound in some media. Note that the speed of sound varies with temperature.

Medium	Speed of sound / m s^{-1}
Carbon dioxide at 0 °C	259
Carbon dioxide at 20 °C	267
Air at 0 °C	330
Air at 20 °C	340
Water at 0 °C	1402
Water at 20 °C	1482
Concrete	5000
Steel	6000

Table 7.2a *Speed of sound through different materials at different temperatures.*

→ This video demonstrates the Phone experiment.

Video

Let us measure the speed of sound in air using two smartphones.

Phone experiment

Speed of sound in air

→ Worksheet is available on OUP web.

What to do

You may search the app on the Internet.

physics phone experiment app

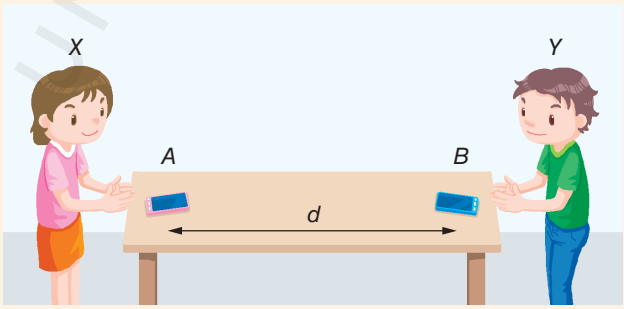
1 Prepare two smartphones *A* and *B* with acoustic stopwatches, which can be started and stopped by sound.

2 Go to a quiet open area. Place the two phones at least 4 m apart and measure the distance *d* between them.

3 Two people *X* and *Y* stand apart with the phones in between. They align their hands and phones along a straight line.

See Example 6 (p.168) for the reason.

4 *X* claps hands once. The clapping sound starts both stopwatches, but the closer stopwatch *A* starts first. Stopwatch *B* starts with a delay of time *t*.



The sound takes time *t* to travel from *A* to *B*.

5 Then *Y* claps hands once. This clap stops the two stopwatches, but *B* stops first and *A* stops with the same delay of time *t*.

6 Record time t_A and t_B shown on *A* and *B*.

$t_A = t_B + 2t$

7 Find the speed of sound v by $t = \frac{t_A - t_B}{2}$ and $v = \frac{d}{t}$.

8 Repeat the experiment using different distances *d*. Take the average value of v.

What you find

What is the measured speed of sound in air? Depend on experimental data

Video

→ This video shows an experiment demonstrating how to measure the speed of sound in air.

Example 6 | Measuring the speed of sound in air

Mary does an experiment to measure the speed of sound in air. She connects two microphones *X* and *Y* to a timer (Fig a). She then hits a metal plate with a hammer to make a sound. When *X* receives the sound, the timer starts. When *Y* receives the sound, the timer stops. The speed of sound can be found from the reading *t* of the timer and the distance *d* between *X* and *Y*.

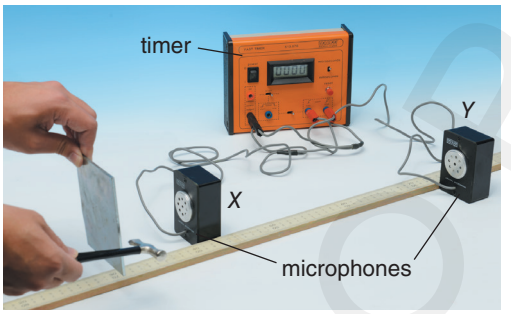


Fig a

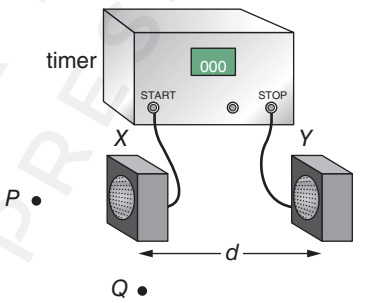


Fig b

(a) At which point in Figure b, *P* or *Q*, should Mary make the sound?

(b) She repeats the experiment with different distances *d* and records the following data.

***d* / m**	0.4	0.5	0.6	0.7
***t* / ms**	0.82	1.47	1.75	2.05

Estimate the speed of sound in air from the data.

(c) The value of *d* should not be too small. Explain.

(d) The metal plate should not touch the bench. Explain.

Solution

(a) *P*

Tip
If she makes the sound at *Q*, the distance travelled by the sound in time *t* will be equal to *QY* – *QX*, not *d*.

(b)

***d* / m**	0.4	0.5	0.6	0.7
***t* / ms**	0.82	1.47	1.75	2.05
***v* / m s^{-1}**	488	340	343	341

Tip
Taking the average value of several sets of data can average out the error in each set of data.
The first data is omitted because it deviates a lot from the other data.

$$\text{The speed of sound} = \frac{340 + 343 + 341}{3} = 341 \text{ m s}^{-1}$$

(c) If *d* is small, the percentage error in measuring *d* will be large.

(d) If the metal plate touches the bench, some sound waves will travel through the bench. Since sound travels much faster in the bench than in air, the experimental results will be affected.

Google Forms Microsoft Forms

Quick check 5

→ E-version in OUP Exercise Platform is also available.

1 *True or false:*

(a) Sound cannot travel through a vacuum. (T/F)

(b) As sound travels through air, it transmits energy and transfers air particles. (T/F)

(c) When sound travels through air, the air particles vibrate along the travelling direction of the sound. (T/F)

2 In which of the following media does sound travel the fastest?

A Air

B Water

C Steel

Learn by practice 1

1 Two microphones *X* and *Y* are 3 m apart. David claps his hands once, with his hands, *X* and *Y* aligning along a straight line. Microphone *Y* detects the clapping sound 8.8 ms after *X* does.

(a) Estimate the speed of sound in air.

(b) David repeats the experiment by hitting a metal plate with a hammer so that a sound with a higher frequency is produced. Will the speed of sound measured be different from that in (a)?

DSE goal

Realize sound can exhibit reflection, refraction, diffraction and interference.

DSE exam

13(1A)Q19, 14(1A)Q18, 15(1A)Q19, 16(1A)Q21, 18(1B)Q7(b), 19(1A)Q15, Q21, 20(1A)Q19, 23(1A)Q19

3 Wave phenomena of sound

Like other waves, sound waves exhibit reflection, refraction, diffraction and interference. The last two phenomena provide evidence that sound is a wave.

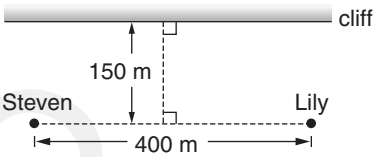
a Reflection

The reflection of sound is commonly observed in daily life. For example, when you shout on top of a mountain, you can hear the sound reflected by other mountains (Fig 7.2g). The reflected sound is called an echo and is softer than the original sound. Many animals, such as bats and whales, make use of the reflection of sound for navigation and hunting.

Fig 7.2g *When you shout on top of a mountain, you can hear an echo.*

Example 7 Echoes

Steven and Lily, 400 m apart, stand 150 m in front of a smooth vertical cliff. Steven shouts once at time $t = 0$. The speed of sound in air is 340 m s^{-1}.



(a) Steven hears the echo at $t = t_1$. Find t_1.

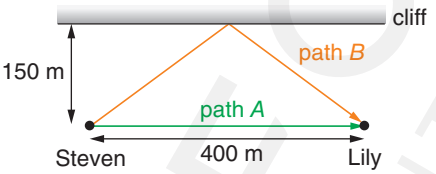
(b) Lily hears the shout directly from Steven and then its echo from the cliff. Find the time interval between the two sounds.

Solution

Tip
The sound travels from Steven to the cliff and back to him.
$\therefore d = 150 \times 2 = 300$ m

(a) $t_1 = \dfrac{d}{v} = \dfrac{150 \times 2}{340} = 0.882$ s

(b) Steven's shout travels to Lily along two paths.



Length of path $B = 2 \times \sqrt{150^2 + \left(\dfrac{400}{2}\right)^2} = 500$ m

Difference between the paths $D = 500 - 400 = 100$ m

Time interval $= \dfrac{D}{v} = \dfrac{100}{340} = 0.294$ s

Google Forms Microsoft Forms

Quick check 6

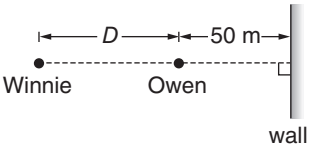
→ E-version in OUP Exercise Platform is also available.

1 Iris shouts in front of a cliff and hears the echo after time t. The distance between Iris and the cliff is d. The speed of sound in air is v. Complete the table.

	t / s	d / m	v / m s^{-1}
(a)	0.5		340
(b)		500	340

Learn by practice 2

1 As shown on the right, Owen shouts at a wall. Winnie hears the echo after 0.5 s. Find the distance D between Winnie and Owen. Take the speed of sound in air as 340 m s^{-1}.



b Refraction

When sound waves travel from one medium to another at an angle, refraction occurs. The direction the sound travels changes due to a change in wave speed.

The set-up shown in Figure 7.2h can be used to demonstrate the refraction of sound. Sound travels slower in carbon dioxide than in air. The balloon filled with carbon dioxide acts like a lens and focuses the sound wave at a point. A loud sound will be detected at this point.

Video

→ This video demonstrates the refraction of sound.

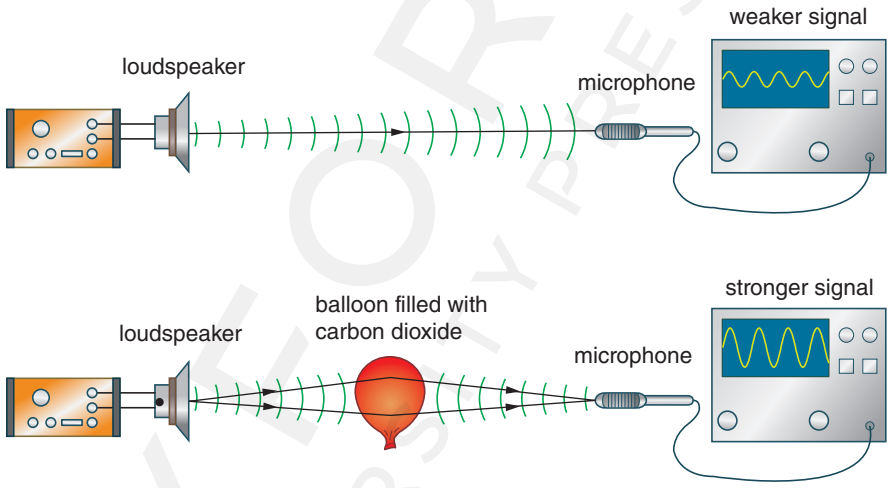


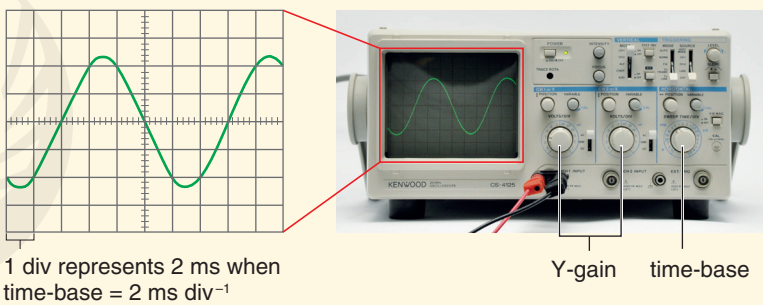
Fig 7.2h *Refraction of sound.*

→ Teaching notes and exercises are available on OUP web.

Skill
Experiment

Cathode-ray oscilloscope (CRO)

A *cathode-ray oscilloscope* (CRO) can provide a visual image of electrical signals. It produces an electron beam that hits the screen to form a bright spot. The beam is deflected horizontally as time passes (the time-base), while the beam is deflected vertically according to the strength and the sign of the electrical signals (the Y-gain). The deflections along the two directions cause the beam to trace a graph of the signal over time.



Cross-link
You can learn more about microphone in Book 4 Ch 7.2.

A microphone connected to a CRO can be used to detect sound. When sound reaches the microphone, it makes a tiny metal plate inside the microphone vibrate. This produces an electrical signal with the same frequency as the sound. The signal is shown on the CRO as a waveform, which is effectively an s–t graph of a particle on the sound wave.

cathode-ray oscilloscope (CRO) 陰極射線示波器

Know more

Loudspeakers

Some loudspeakers contain speaker cones in different sizes. Do you know why?

▸ p.198

Video

→ This video shows an experiment demonstrating the diffraction of sound.

C Diffraction

We can often hear people talking behind a doorway (Fig 7.2i). This is because sound undergoes diffraction after passing through the door. The diffraction is noticeable as the typical wavelength of a human voice is comparable to the width of a doorway.

We can use a loudspeaker and a long paper tube to show the diffraction of sound (Fig 7.2j). Sound can be heard at different positions in front of the paper tube due to diffraction.

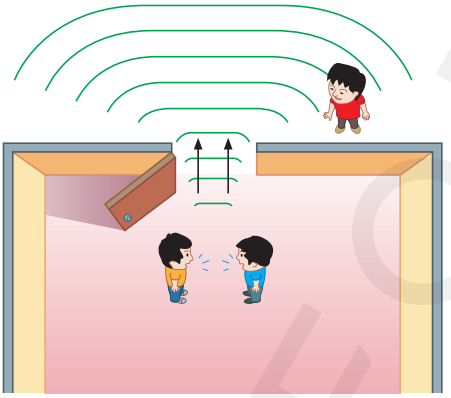


Fig 7.2i *We can hear people talking behind a doorway.*

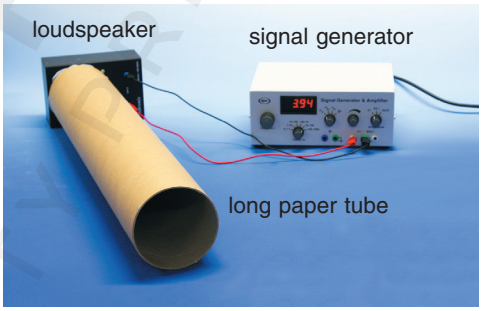


Fig 7.2j *A set-up demonstrating the diffraction of sound.*

Google Forms Microsoft Forms

Quick check 7

→ E-version in OUP Exercise Platform is also available.

1 A sound wave travels from concrete to air (Fig a). It travels faster in concrete than in air.

(a) How does the frequency of the sound wave change?

(b) How does the wavelength of the sound wave change?

(c) Which path, *X*, *Y* or *Z*, is the possible path of the sound wave in air?

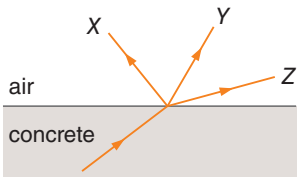


Fig a

2 A sound wave diffracts after passing through a tube (Fig b).

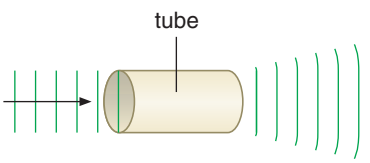


Fig b

Which of the following methods can increase the degree of diffraction of the sound wave? Put a tick in the corresponding box(es).

☐ Reduce the frequency of the sound wave.

☐ Replace the tube with another one with a smaller opening.

☐ Put the whole set-up in water.

d Interference

We can demonstrate the interference of sound using two smartphones.

→ This video demonstrates the Phone experiment.

Video

Interference of sound

Phone experiment

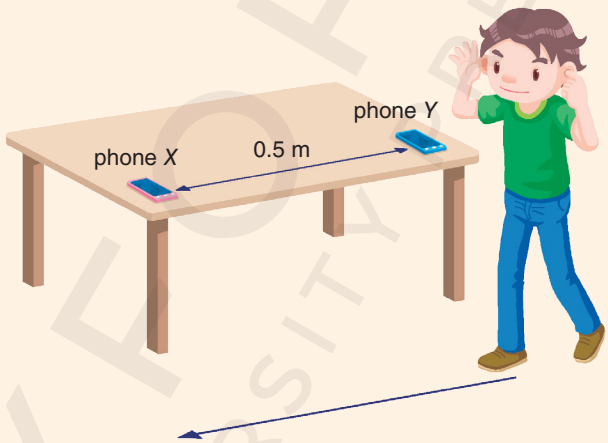
→ Worksheet is available on OUP web.

What to do

You may search the app on the Internet.

physics phone experiment app

1 Prepare two smartphones *X* and *Y* with tone generators. Use them to produce 2000-Hz sound at about the same loudness.
2 Place the phones about 0.5 m apart on a table.
3 Cover one ear and walk across the area in front of the phones. Note any change in the loudness of the sound.



The two phones are coherent sources.

You may search the app on the Internet.

sound meter app

4 Use a third smartphone *Z* as the detector. Open an app that can detect the loudness of sound. Hold phone *Z* and walk across the area in front of phones *X* and *Y* on the table. Note any change in the loudness of the sound.

What you find

Is the loudness of the sound constant?

In the activity, alternate loud and soft sounds are detected across the area in front of the phones. This can be explained by the interference of sound waves. Constructive interference (loud sound) occurs at some positions and destructive interference (soft sound) occurs at some other positions. As a result, alternate loud and soft sounds are heard (Fig 7.2k).

Soft sounds can be heard (instead of silence) at positions of destructive interference. One of the reasons for this is the reflection of sound from the surrounding environment.

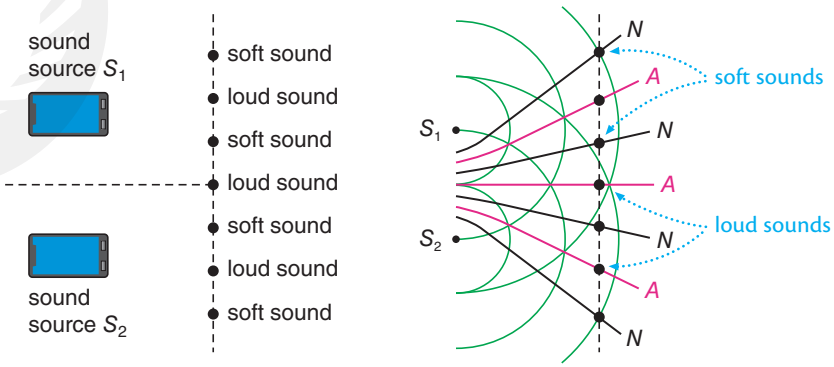


Fig 7.2k *Interference of sound.*

Now we have seen that sound exhibits the unique phenomena of waves, i.e. diffraction and interference. This shows that sound is a wave.

Sound exhibits reflection, refraction, diffraction and interference.

The nodal lines and antinodal lines can be located when many people participate in this experiment by all standing at positions of loud sound / low sound).

Video

→ This video demonstrates interference of sound and shows the nodal lines of the interference pattern.

Know more

Noise-cancelling headphones

Noise-cancelling headphones cancel out ambient noise by making use of the interference of sound.

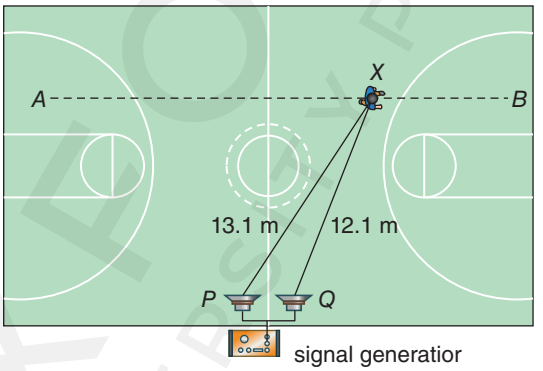
▸ p.198

Tip

The separation between antinodal lines decreases as the wavelength decreases.

Example 8 Interference of sound

Two loudspeakers P and Q are connected to a signal generator in a basketball court. The sounds that P and Q produce have a frequency of 850 Hz and are in phase. Tom walks along AB. He stops at point X which is 13.1 m from P and 12.1 m from Q. The speed of sound in air is 340 m s^{-1}.



(a) Find the wavelength of the 850-Hz sound.

(b) Explain whether Tom hears a loud sound or a soft sound at X.

(c) State an advantage of carrying out this experiment in an outdoor area rather than inside a laboratory.

(d) How will the separation between two consecutive loud sounds along AB change if the frequency of the sound increases?

Solution

(a) By $v = f\lambda$,

$$\lambda = \frac{v}{f} = \frac{340}{850} = 0.4 \text{ m}$$

(b) Path difference at X = 13.1 m – 12.1 m = 1.0 m = 2.5λ

$\therefore$ Destructive interference occurs and Tom hears a soft sound.

(c) It reduces the reflection of sound from the surrounding environment.

(d) If the frequency increases, the wavelength will decrease.

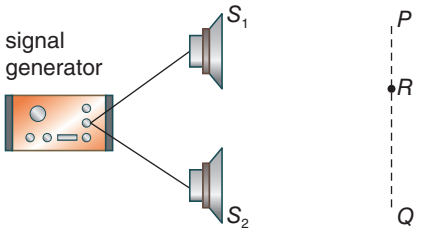
Therefore, the separation between two consecutive loud sounds along AB will decrease.

Quick check 8

→ E-version in OUP Exercise Platform is also available.

Google Forms | Microsoft Forms

1 Two loudspeakers S_1 and S_2 are connected to a signal generator. Alternate loud and soft sounds are heard along PQ.



(a) How can we increase the separation between two consecutive loud sounds along PQ? Put a tick in the corresponding box(es).

- ☐ Increase the wavelength of the sound.
- ☐ Increase the separation between S_1 and S_2.

(b) R is 1.2 m from S_1 and 1.56 m from S_2. The wavelength of the sound is 0.24 m. Is a loud or soft sound heard at R?

☆ HKDSE 2016 Paper 1A Q21

→ This video demonstrates how this question is solved.

Video

DSE practice 1 | Interference pattern of sound

P and Q are two loudspeakers producing sound waves in phase. A loud sound is detected at point M, which is 5.4 m from P and 5.6 m from Q. Point O is equidistance from P and Q. Which of the following statements must be correct?

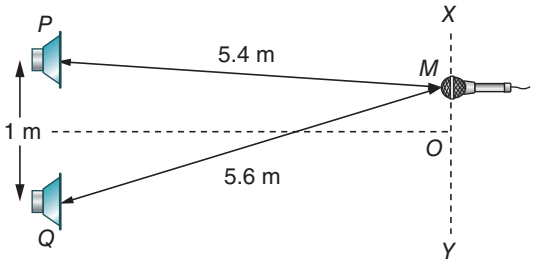
(1) The wavelength of the sound waves is 20 cm.

(2) The fringe separation equation $\Delta y = \frac{\lambda D}{a}$ cannot be used in this case because the equation can only be applied to light.

(3) If the sound waves from P and Q are in antiphase, a soft sound will be detected at M.

A (1) only B (3) only

C (1) and (2) only D (1), (2) and (3)



Solution

We do not know the number of loud sounds between O and M, so we cannot find the wavelength from the information given.

∴ (1) is not necessarily correct.

The equation $\Delta y = \frac{\lambda D}{a}$ can be applied only when $\Delta y \ll D$ and $a \ll D$.

Since both conditions are not satisfied, the equation cannot be applied.

∴ (2) is incorrect.

If the sound waves from P and Q are in antiphase, they will arrive M in antiphase, so destructive interference occurs.

∴ (3) is correct.

∴ The answer is B.

Common mistake ⚠
Students may not know the two conditions required to apply the equation. Some wrongly think that the equation can only be applied to light.

Common mistake ⚠
Students may not know how to find the type of interference at a certain position when the sources are in antiphase.

▶ Revision 7 Q13 (p.192)

Exercise 7.2

Take the speed of sound in air as 340 m s^{-1}.

*1 L_1 and L_2 are two identical loudspeakers (Fig a). They emit sound waves in phase. Loud sounds are heard at P and Q, where P is equidistance from L_1 and L_2. No more loud sound is heard between P and Q.

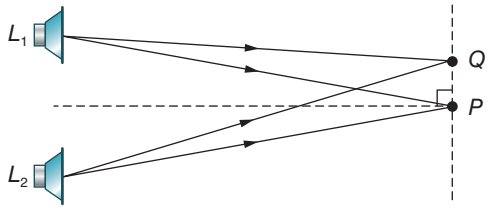


Fig a

If L_1Q is 4.9 m and L_2Q is 5.3 m, what is the frequency of the sound?

A 425 Hz B 850 Hz
C 1275 Hz D 1700 Hz

*2 A boy stands between two cliffs P and Q (Fig b) and makes a sound towards P at time t = 0. He hears the first echo at t = 0.8 s and the second echo at t = 1.68 s. Assume the sound does not travel from the boy to Q directly. Find the distance between P and Q.

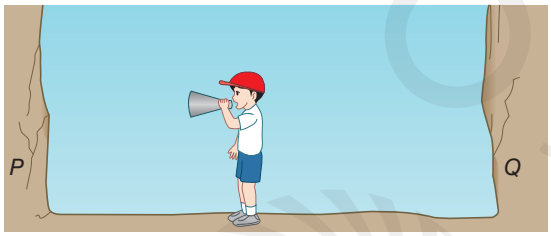


Fig b

A 136 m B 150 m
C 286 m D 340 m

*3 It is known that sound waves travel faster in hot air than in cold air. When a sound wave enters the region of cold air as shown (Fig c), how will its direction of travel and wavelength change?

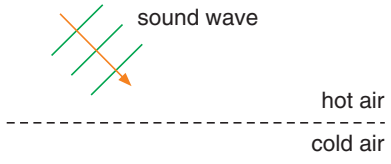


Fig c

	Direction of travel	Wavelength
A	bend towards normal	increase
B	bend away from normal	increase
C	bend towards normal	decrease
D	bend away from normal	decrease

*4 A mosquito flaps its wings 600 times per second to produce its annoying sound (Fig d).

Fig d

(a) What is the frequency of the sound waves it produces?

(b) How far does the sound travel between two consecutive flaps?

(c) Explain how sound is produced when a mosquito flaps its wings.

*5 It is known that the wavelength of a sound wave in air is 1.33 m. When this sound enters medium X, its wavelength becomes 5.47 m.

(a) Find the frequency of this sound in air.

(b) Find the speed of this sound wave in medium X.

*6 A loudspeaker is placed above the floor as shown (Fig e). When a microphone is moved along XY, alternating loud and soft sounds are detected. Explain why this happens.

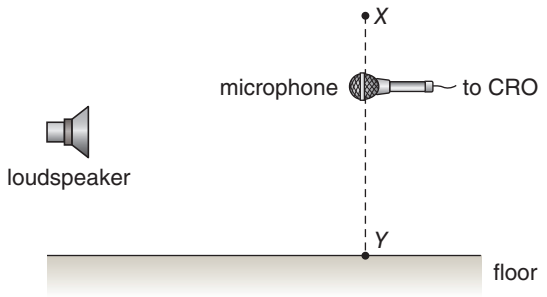


Fig e

7.3 More about sound

Bats (Fig 7.3a) can locate and identify objects by emitting sound and detecting the echoes. However, we may not hear this sound even if we are very close to a bat. Why?

Fig 7.3a A bat.

DSE goal

Determine the audible frequency range.

Examine the existence of ultrasound beyond the audible frequency range.

DSE exam

12(1A)Q23, 14(1A)Q17, 16(1A)Q23, 20(1A)Q21, 21(1A)Q20, 22(1A)Q22

Flipped classroom

Watch a video and answer the questions.

1 Audible sound and ultrasound

a Audible frequency range

Humans only hear sounds of frequencies from about 20 Hz to 20 kHz. (Fig 7.3b). This range of frequency is called the **audible frequency range**. Sound within this range is called audible sound.

In air of 20 °C, sound travels at 340 m s^{-1}. By $v = f\lambda$, the wavelength of audible sound ranges from 0.017 m to 17 m.

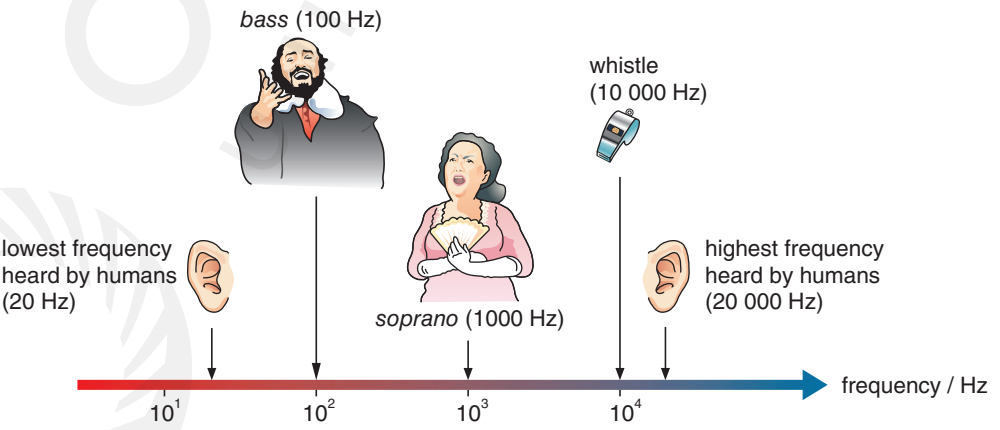


Fig 7.3b The audible frequency range.

The range varies from person to person and narrows with age.

You can use an online tune generator to determine your audible frequency range. Remember to keep the volume low.

Useful link

Audible frequency range: 20 Hz ⩽ f ⩽ 20 000 Hz

The sounds emitted by bat can have a frequency up to 200 kHz, so we cannot hear the sound.

audible frequency range 聽頻範圍　　bass 男低音歌手　　soprano 女高音歌手

b Ultrasound

Ultrasound can also be called ultrasonic waves.

Sound with a frequency lower that 20 Hz is called infrasound.

▶ Sound with a frequency higher than 20 kHz is called **ultrasound**. It travels at the **same speed** as audible sound in air.

We cannot hear ultrasound but many animals can (Fig 7.3c).

20–20 000 Hz

55–77 000 Hz

64–44 000 Hz

150–150 000 Hz

Fig 7.3c *Audible frequency ranges of some animals.*

Frequency of ultrasound > 20 000 Hz

Cross-link

The working principle of ultrasound scanning will be further discussed in Book E4 Ch 2.

▶ Ultrasound can be used in a *sonar* system, which locates an object by emitting ultrasound and detecting the echoes from the object. Sonar can be used to detect fish (Fig 7.3d). Ultrasound scanning (Fig 7.3e) and ultrasonic flaw detectors (Fig 7.3f) also make use of sonar.

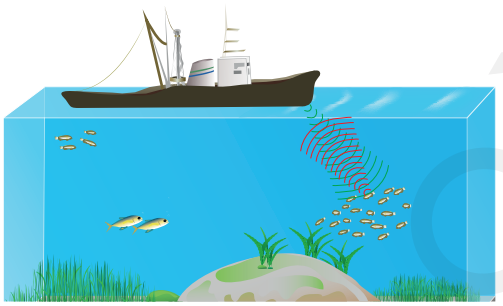


Fig 7.3d *Using sonar to detect fish.*

Fig 7.3e *Ultrasound scanning.*

Fig 7.3f *Ultrasonic flaw detectors.*

Ultrasound can also be used to clean small and delicate objects like jewellery and spectacles (Fig 7.3g). When ultrasound is focused on a small region, a large amount of energy is transmitted to the region. This technique is commonly used to break large *kidney stones* (Fig 7.3h).

Fig 7.3g *Ultrasound cleaner.*

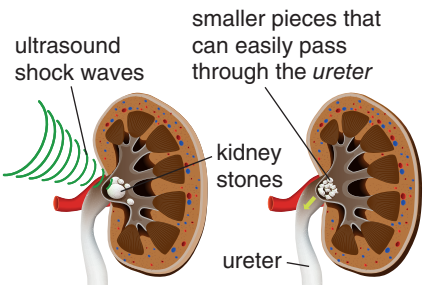


Fig 7.3h *Using ultrasound to break a kidney stone.*

ultrasound 超聲波　　sonar 聲納　　kidney stones 腎石　　ureter 輸尿管

Google Forms Microsoft Forms

Quick check 9

→ E-version in OUP Exercise Platform is also available.

1 *True or false:*

(a) Ultrasound travels faster than audible sound in air. (T/F)

(b) Ultrasound has a shorter wavelength than audible sound. (T/F)

(c) Ultrasound can travel through a vacuum. (T/F)

(d) When ultrasound travels through a small gap, it diffracts less than audible sound. (T/F)

☆ HKDSE 2014 Paper 1A Q17

→ This video demonstrates how this question is solved.

Video

DSE practice 2 Sonar

A submarine moves at a uniform speed of 16 m s^{-1} towards an underwater cliff (Fig a). It emits an ultrasound pulse and receives the echo reflected by the cliff at the time instants shown in Figure b. The ultrasound pulse travels in seawater at 1500 m s^{-1}.

Fig a

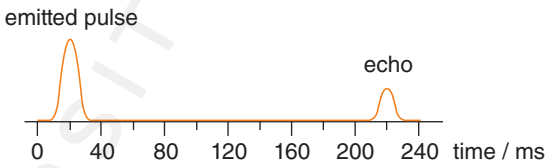


Fig b

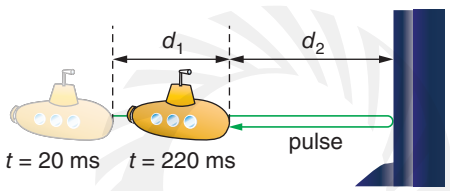
(a) How far is the submarine away from the cliff when it receives the echo? (3 marks)

(b) Why cannot microwaves be used to detect objects underwater? (1 mark)

Solution

(a) Time t between emitting and receiving the pulse

$= 220 - 20 = 200$ ms 1M



Distance d_1 travelled by submarine $= v_s t = (16)(200 \times 10^{-3}) = 3.2$ m

Common mistake
Students may miss out the distance travelled by the submarine during the period from emitting to receiving the pulse.

Distance travelled by pulse $= v_p t = d_1 + 2d_2$

$(1500)(200 \times 10^{-3}) = 3.2 + 2d_2$ 1M

$d_2 = 148$ m 1A

The submarine is 148 m from the cliff when it receives the echo.

(b) This is because microwaves are easily absorbed by water. 1A

Revision 7 Q12 (p.192)

DSE goal

Compare general properties of sound waves and light waves.

DSE exam

12(1A)Q19, 17(1A)Q21, 19(1A)Q17

2 Sound and light

Sound and light allow us to sense the world. Both are waves, but they have some different properties.

1 Sound is a longitudinal wave. Light is a transverse wave.

2 Sound is a mechanical wave. It cannot travel through a vacuum. Light is an electromagnetic wave. It can travel through a vacuum.

3 Sound travels at 340 m s^{-1} in air. It travels **faster** in liquids and solids. Light travels at 3×10^8 m s^{-1} in air. It travels **slower** in liquids and solids. Figure 7.3i shows how they behave when travelling from air to water.

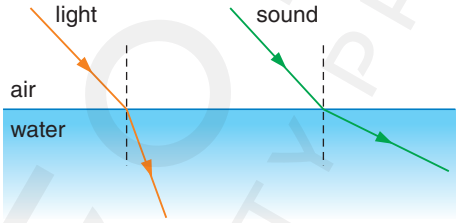


Fig 7.3i *Light and sound entering water from air.*

4 In air, the wavelength of audible sound ranges from about 0.017 m to 17 m, while the wavelength of visible light ranges from about 400 nm to 700 nm. In daily life, the diffraction of audible sound is significant while that of visible light is negligible.

5 The frequency of audible sound ranges from about 20 Hz to 20 000 Hz. The frequency of visible light ranges from about 4.3×10^{14} Hz to 7.5×10^{14} Hz.

Table 7.3a summarizes the differences between sound and light.

	Sound wave	Light wave
Nature	mechanical wave	electromagnetic wave
Type of wave	longitudinal wave	transverse wave
Travelling medium	solids, liquids and gases (but not vacuums)	transparent or translucent solids, liquids and gases, and also vacuums
Travelling speed in air	340 m s^{-1}	3×10^8 m s^{-1}
Range of wavelengths detectable by human bodies in air	0.017 m to 17 m	400 nm to 700 nm
Range of frequencies detectable by human bodies	20 Hz to 20 000 Hz	4.3×10^{14} Hz to 7.5×10^{14} Hz

Table 7.3a *Differences between light and sound.*

Quick check 10

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

(For Q1–2.) A girl sees a flash of lightning in the sky. After 4 s, she hears the bang.

1 Why does the bang come after the flash of lightning?

A Light has a higher frequency than sound.

B Light travels faster than sound in air.

2 How far is the thunderstorm away from the girl? The speed of sound in air is 340 m s^{-1}.

A 1360 m B 1.2×10^9 m

3 *True or false:*

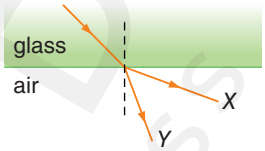
(a) Both ultrasound and microwaves travel at 3×10^8 m s^{-1} in air. (T/F)

(b) Ultrasound is a longitudinal wave while microwave are a transverse wave. (T/F)

4 A wave travels from glass to air. Which path, *X* or *Y*, shows the possible path of the wave in air

(a) if the wave is a light wave;

(b) if the wave is a sound wave?



5 Jeff stands next to the door of a room. He can hear people inside talking but cannot see the people. Which of the following is the reason?

A Sound is a mechanical wave while light is an electromagnetic wave.

B The wavelength of audible sound is much longer than that of visible light.

Flipped classroom

Watch a video and answer the questions.

3 Musical notes and noise

The sound from a piano is pleasant to hear but the sound from a *jackhammer* is not (Fig 7.3j). What is the difference?

(i) (ii)

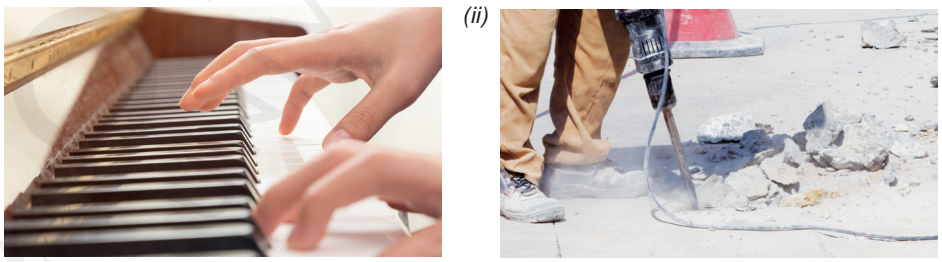


Fig 7.3j *(i) A piano and (ii) a jackhammer produce different sounds.*

The sound from a piano is called a **musical note** (or a note). A note is caused by regular vibrations. The sound from a jackhammer is called a **noise**. A noise is caused by irregular vibrations. Figure 7.3k shows the waveforms of a note and a noise. They are effectively *s*–*t* graphs of particles on the sound waves.

(i) (ii)

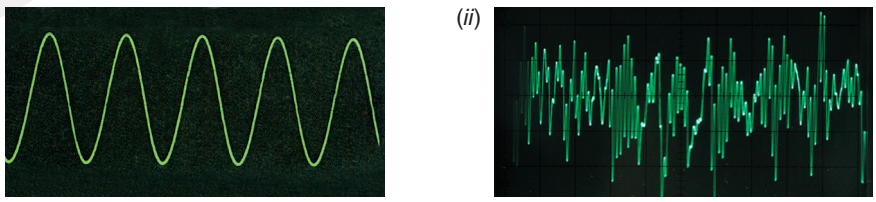


Fig 7.3k *Waveforms of (i) a note and (ii) a noise.*

jackhammer 風砲 musical note 樂音 noise 噪音

→ The video 'Musical notes (1)' shows an experiment studying notes using a CRO.

→ The video 'Musical notes (2)' shows an experiment studying notes using a sound sensor and a data-logger.

DSE goal

Compare musical notes using pitch, loudness and quality.

Relate frequency and amplitude with pitch and loudness of a note.

DSE exam

12(1A)Q22, 17(1A)Q18

Experiments studying notes:

Video

There are many apps that can show the waveforms of sound.

Useful link

iOS Android

Simulation

→ This simulation generates musical notes at different frequencies produced by different musical instruments, and show the corresponding waveforms.

a Describing notes

When we compare musical notes, we may compare their **pitch**, **loudness** and **quality**.

The pitch of a note refers to how high or low the note is. The higher the frequency f of the sound wave, the higher the pitch of the note. Figure 7.3l shows the waveforms of some notes, with the scale in the amplitude and time axes being the same.

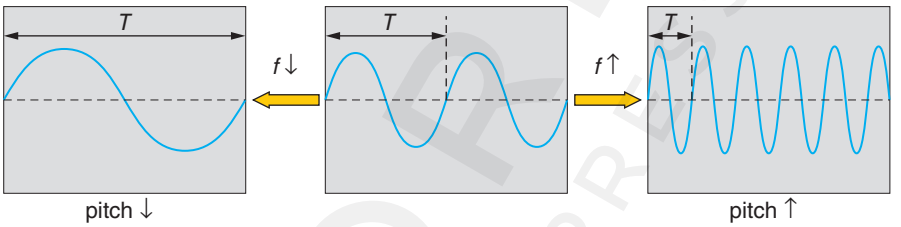


Fig 7.3l Notes of different pitches.

By $f = \frac{1}{T}$, $f \uparrow \Rightarrow T \downarrow$

Pitch is the subjective feeling about the highness or lowness of a note.

The loudness of a note is related to the amplitude of the sound wave. The larger the amplitude, the louder the note (Fig 7.3m).

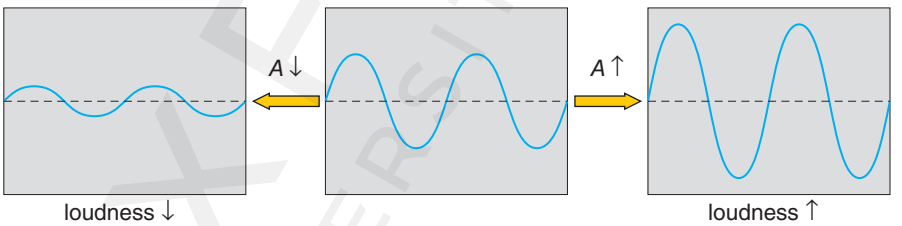


Fig 7.3m Notes of different loudness.

Loudness is the subjective feeling about the 'strength' of a note.

Notes produced by different musical instruments sound differently. They are said to have different qualities. In Figure 7.3n, notes (all with the same frequency) produced by a tuning fork, a piano and a flute are shown. We can see that they have **different waveforms**. This results in the different qualities perceived by audience.

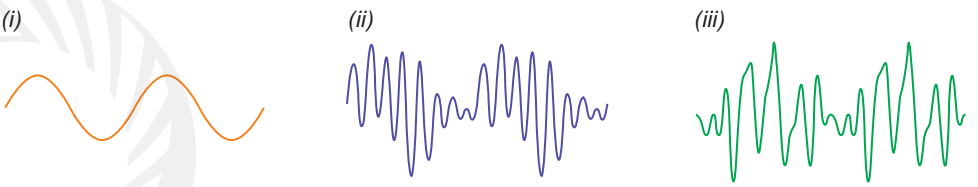


Fig 7.3n The middle C sounded by (i) a turning fork, (ii) a piano and (iii) a flute.

Know more

Quality of sound

A note is composed of a combination of frequencies.

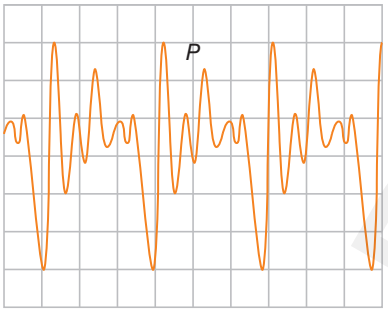
▸ p.198

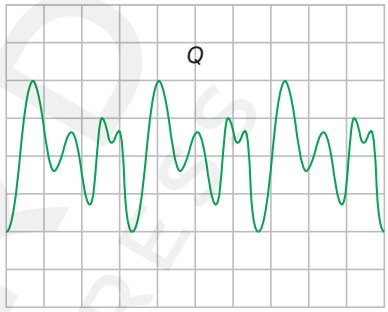
The pitch of a note increases with the frequency of the sound wave.
The loudness of a note increases with the amplitude of the sound wave.
Notes of different qualities have different waveforms.

pitch 音調 loudness 響度 quality 音品

Example 9 Musical notes

The waveforms of notes P and Q are as shown, with the same scales in the time and amplitude axes.





(a) Compare the loudness of these two notes.

(b) Compare the pitch of these two notes.

(c) Do these two notes have the same quality?

Solution

(a) P is louder than Q.

(b) P has a higher pitch than Q.

(c) No, they have different qualities.

Tip

(a) The amplitude of P is larger.

(b) The period of P is shorter, so its frequency is higher.

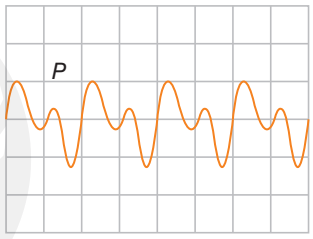
(c) The waveforms are different.

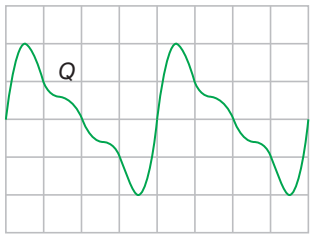
Google Forms Microsoft Forms

Quick check 11

→ E-version in OUP Exercise Platform is also available.

1 The waveforms of notes P and Q are as shown, with the same scales in the time and amplitude axes.





Determine whether the following statements are true.

(a) P has a higher pitch than Q. (T/F)

(b) P is louder than Q. (T/F)

(c) P and Q are produced by different musical instruments. (T/F)

DSE goal

Represent sound intensity level using the unit decibel.

DSE exam

18(1A)Q21

Cross-link

Sound intensity level will be further discussed in Book E4 Ch 2.1.

b Sound intensity level

In daily life, instead of amplitude, **sound intensity level** is used to describe the loudness of sound. This quantity is measured in **decibels (dB)**. Table 7.3b shows some typical sound intensity levels.

Threshold of hearing	0 dB
Country park	20 dB
Quiet talking	40 dB
Normal talking	60 dB
Vacuum cleaner at 3 m	70 dB
Road drilling at 5 m	100 dB
Rock concert	120 dB
Aeroplane take-off at 30 m	140 dB

Table 7.3b *Some sound intensity levels.*

iOS Android

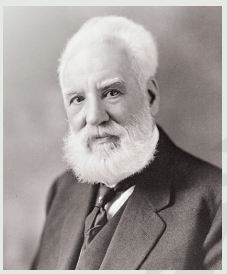
There are many apps that can measure the sound intensity levels.

Useful link

The zero point, i.e. 0 dB, is called the *threshold of hearing*. It is not 'no sound' but rather the lowest sound intensity level that can be heard by the human ear. The sound intensity level of a sound can be lower than 0 dB, but the human ear cannot hear this sound.

Know more

Alexander Bell (1847–1922)



Alexander Bell is the inventor of the telephone. He also spent most of his life teaching the deaf to speak and researching sound production. The units bel and decibel are named after him.

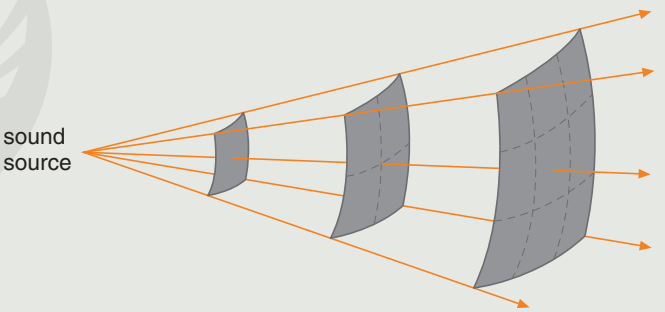
OUP eResources

Intensity of waves

Intensity is an important quantity in studying waves. It indicates the rate at which energy is transmitted to a place. It is defined as the power per unit area:

$$\text{Intensity} = \frac{\text{power}}{\text{area}} = \frac{\text{energy}}{\text{area} \times \text{time}} \left(I = \frac{P}{A} = \frac{E}{At} \right)$$

Take sound as an example. As the sound from a **point source** travels outwards, its power remains unchanged but the area that the sound passes through increases. Therefore, the further the observer is from the source, the lower the intensity detected. This is why you hear a louder sound when the observer is you move closer to a sound source.



Sound intensity level L is defined by comparing the sound intensity I with a reference level I_0, according to the formula $L = 10 \log \frac{I}{I_0}$.

sound intensity level 聲強級　decibel (dB) 分貝　threshold of hearing 聽覺閾
Alexander Bell 貝爾

DSE goal

Discuss the effects of noise pollution and the importance of acoustic protection.

C Noise pollution

Loud sound, whether it is a note or a noise, can be harmful. Very loud sound (above 140 dB) causes permanent damage to the ear. Sound at 130 dB produces ear pain. Constant exposure to loud sound below 130 dB may cause loss of hearing. Besides damage to hearing, living and working in a noisy place may make you feel irritated and stressed.

We should protect our hearing. For example, when you are listening to music with earphones, do not turn the volume too high (Fig 7.3o). The music can be very loud without you realizing it. People working in noisy workplaces should wear ear protectors (Fig 7.3p).

Some earphones can emit music over 120 dB! ▶

Fig 7.3o *Do not turn the volume of music too high.*

Fig 7.3p *Ear protectors prevent damage to hearing due to the noise at work.*

Other measures introduced to control noise pollution in Hong Kong:

(a) Resurface roads with low-noise surfacing materials.

(b) Newly-registered motor vehicles are required to meet noise standards.

Traffic and construction work are two major sources of noise in Hong Kong. The government has introduced some measures to reduce the impact of the noise produced by these sources (Fig 7.3q and 7.3r).

Fig 7.3q *Noise barriers are built along roads and highways to reflect and absorb noise.*

Fig 7.3r *Time controls and noise standards are set for construction work.*

Google Forms

Microsoft Forms

Quick check 12

→ E-version in OUP Exercise Platform is also available.

1 *True or false:*

(a) There is no sound in a room of 0 dB. (T/F)

(b) The sound of a normal conversation is about 20 dB. (T/F)

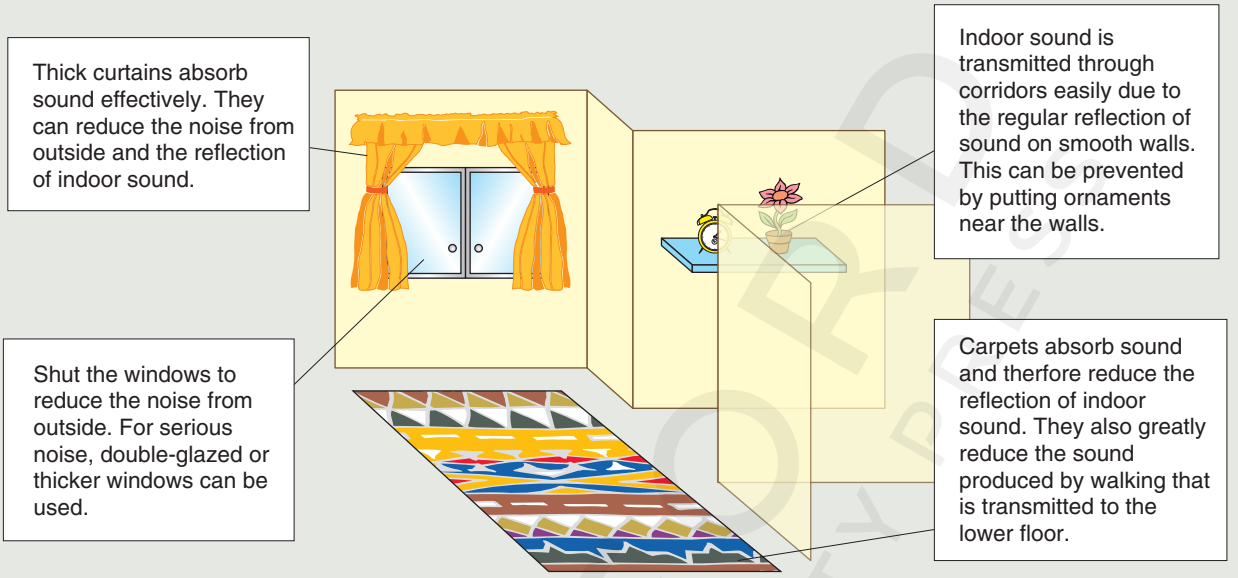
(c) When an aeroplane takes off 30 m away from you, you hear sound of 100 dB. (T/F)

(d) Sound over 140 dB can damage ears. (T/F)

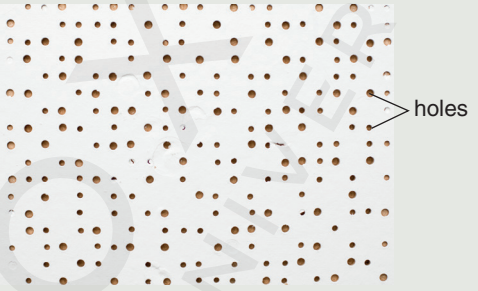
Know more

Interior designs for reducing noise

The figure below shows some ways to reduce the noise in a room.



Moreover, an acoustic board is also commonly used to reduce noise. It is porous with sound absorbing materials embedded inside. When sound waves enter the holes, they undergo multiple reflections. Most of the energy carried by the sound waves is lost during the reflections.



STSE

Noise pollution in Hong Kong

The biggest source of noise pollution in Hong Kong comes from traffic near residential buildings. It is estimated that hundreds of existing roads generate noise louder than 70 dB, and about one million people in Hong Kong are exposed to excessive traffic noise.

Hong Kong people are also familiar with the noise from construction work (e.g. piling, building demolition and renovation), aircraft, ventilation systems in factories, and even neighbours.

The Environmental Protection Department (EPD) has introduced different measures to control noise pollution in Hong Kong. Check them out by visiting the EPD website.

Useful link

Exercise 7.3

(For Q1–2.) The waveforms of musical notes *X*, *Y* and *Z* are as shown, with the same scales in the time and amplitude axes (Fig a).

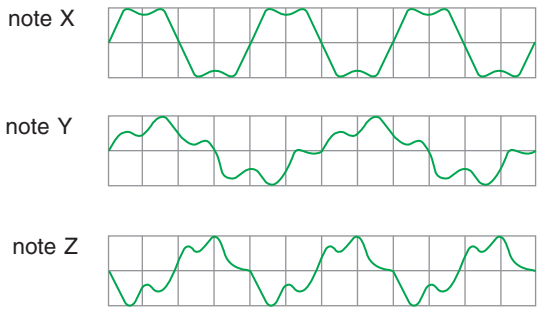


Fig a

1 Which of these notes have the same pitch?

A *X* and *Y* only B *X* and *Z* only
C *Y* and *Z* only D *X*, *Y* and *Z*

2 Which of these notes have the same loudness?

A *X* and *Y* only B *X* and *Z* only
C *Y* and *Z* only D *X*, *Y* and *Z*

★**3** Which of the following statements about sound and light is/are correct?

(1) Both sound and light cannot travel through a vacuum.
(2) Light travels much faster than sound in air.
(3) Sound is a longitudinal wave while light is a transverse wave.

A (1) only B (2) only
C (2) and (3) only D (1), (2) and (3)

★**4** Which of the following statements about audible sound and ultrasound is/are correct?

(1) Audible sound has a higher frequency than ultrasound in air.
(2) Audible sound has a longer wavelength than ultrasound in air.
(3) Audible sound has a lower speed than ultrasound in air.

A (2) only
B (3) only
C (1) and (3) only
D (2) and (3) only

★**5** Which of the following statements is/are correct?

(1) Sound intensity levels are measured in decibels.
(2) The threshold of hearing is at 0 dB.
(3) The sound intensity level cannot be lower than 0 dB.

A (1) only B (1) and (2) only
C (2) and (3) only D (1), (2) and (3)

★**6** Noise barriers built along highways can shield residential buildings from the noise produced by vehicles (Fig b).

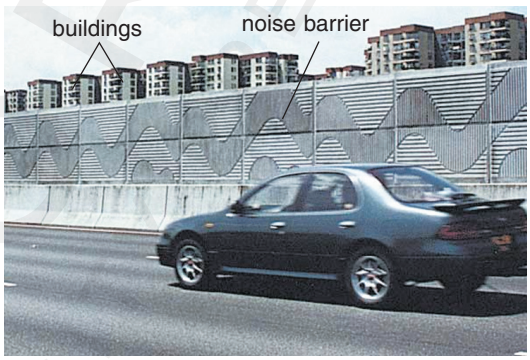


Fig b

(a) Explain how noise barriers reduce the amount of noise reaching the buildings.
(b) Explain why shielding is more effective on high frequency noises than low frequency noises.

★**7** A sonar on a ship can emit sound waves with a frequency of 150 kHz. The speed of sound in seawater is 1500 m s^{-1}.

(a) Explain whether the sound waves emitted are ultrasound.
(b) The ship emits a sound pulse towards the seabed and an echo from the seabed is received (Fig c). The emitted pulse and the echo are both displayed on the screen of a CRO (Fig d). How deep is the sea? (1 ms = 10^{-3} s)

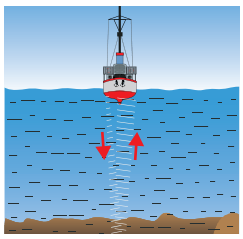


Fig c

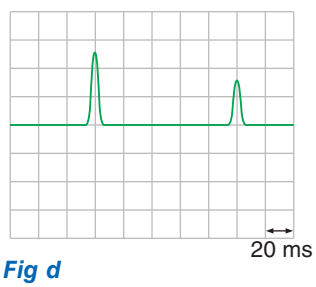
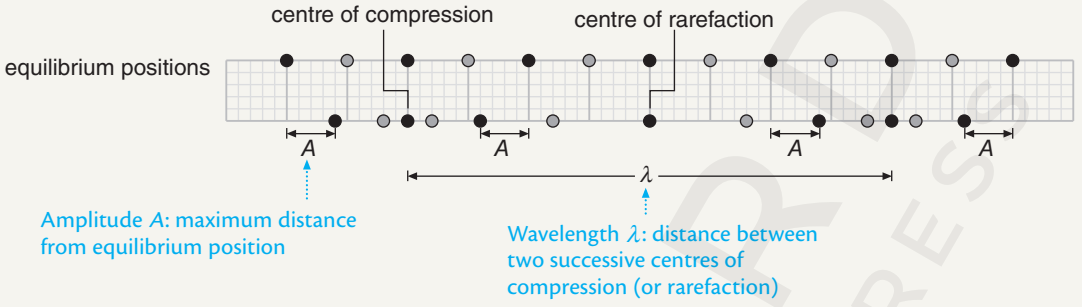


Fig d

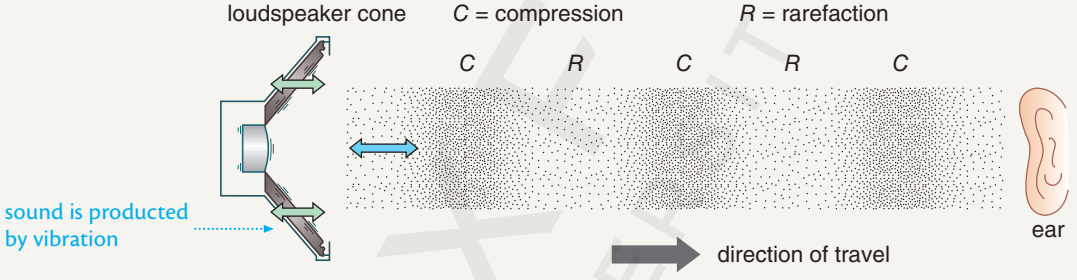
Key concepts 7

1 Longitudinal waves

Q31 Exam Report: Candidates performed well in (a)(i). Only a few candidates did not realise that the percentage error in measurement was being referred to and they answered 'easier to locate the positions of maxima as the dots are brighter'. Some candidates failed to find the correct values of θ or mistook the order of diffraction as $n = 4$ in (a)(iii). Weaker ones wrongly applied $\Delta y = \frac{\lambda D}{a}$ in this situation. In (b)(i), not many candidates knew that the equation $\Delta y = \frac{\lambda D}{a}$ could only be applied when $\lambda \ll a$ and $a \ll D$. Some even thought that the equation was only applicable for light.



2 Sound



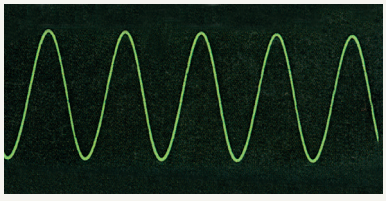
- A longitudinal wave; exhibits reflection, refraction, diffraction and interference
- A mechanical wave; can travel through gases, liquids and solids, but not a vacuum
- Speed of sound depends on the medium it travels through ($v_{solid} > v_{liquid} > v_{gas}$)
- Sound of frequency > 20 000 kHz ⇒ ultrasound
- Comparison of sound waves and light waves:

	Sound wave	Light wave
Nature	mechanical wave	electromagnetic wave
Type of wave	longitudinal wave	transverse wave
Travelling medium	solids, liquids and gases (but not vacuums)	transparent or translucent solids, liquids and gases, and also vacuums
Travelling speed in air	340 m s^{-1}	3×10^8 m s^{-1}
Range of wavelengths detectable by humans in air	0.017 m to 17 m	400 nm to 700 nm
Range of frequencies detectable by humans	20 Hz to 20 000 Hz	4.3×10^{14} Hz to 7.5×10^{14} Hz

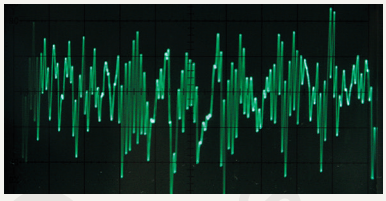
Q32 Exam Report: This question tested candidates' knowledge and understanding in an experiment to find the speed of sound in air. In general, candidates' performance was fair. In (a), quite a number of candidates did not understand how the experiment worked and thus failed to indicate the appropriate location where a sharp loud sound should be produced. Many candidates obtained full marks in (b)(i). A few took the average value of all the four timer readings. Weaker candidates confused microphones with loudspeakers and thought that it was an experiment about the interference of sound, thus employing the wave equation $v = f\lambda$. About half of the candidates failed to score in (b)(ii). Many answers were in fact not adjustments to the experimental setting, e.g. 'repeat the experiment and take average', 'surround the set-up with sound-proof materials', etc.

3 Musical notes and noise

- A musical note is produced by regular vibrations while a noise is produced by irregular vibrations.

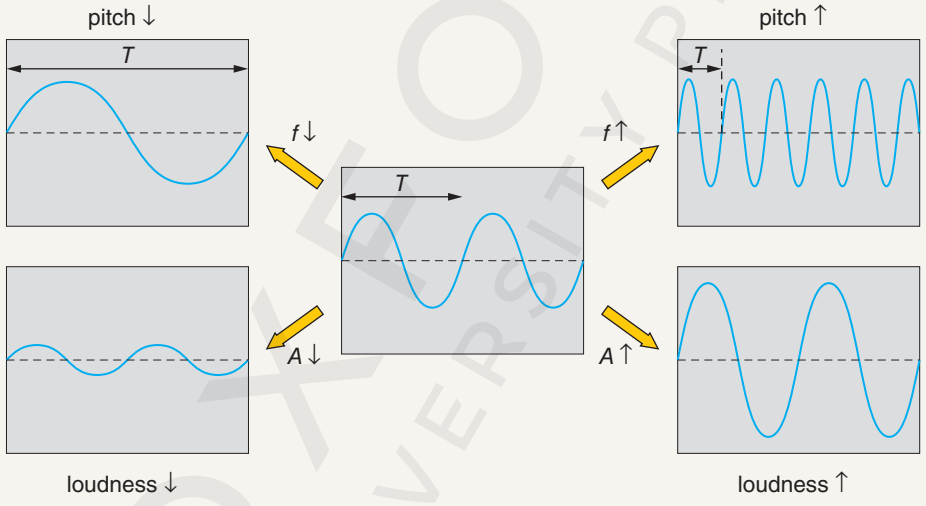


musical note



noise

- The loudness and pitch of a musical note change as its amplitude A and frequency f change.



- Notes of different qualities have different waveforms.

- The loudness of sound is described in terms of sound intensity level. Its unit is decibels (dB).

Key terms

1	audible frequency range 聽頻範圍	p.177
2	decibel (dB) 分貝	p.184
3	loudness 響度	p.182
4	musical note 樂音	p.181
5	noise 噪音	p.181
6	pitch 音調	p.182
7	quality 音品	p.182
8	sound intensity level 聲強級	p.184
9	ultrasound 超聲波	p.178

Revision 7

Multiple-choice questions

★1 7.1 A longitudinal wave travels towards the right through air. The figure below shows the positions of a series of air particles at a certain moment (Fig a).

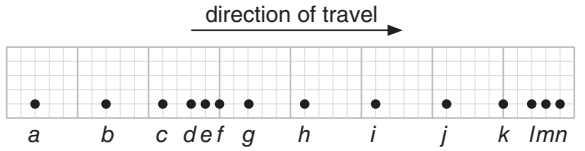


Fig a

Which of the following statements about the particles are correct?

(1) Particle *e* is at the centre of rarefaction.

(2) Particles *d* and *l* are vibrating in phase.

(3) All particles vibrate with the same amplitude.

A (1) and (2) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

★2 7.3 Which of the following statements about sound intensity level are correct?

(1) With its sound intensity level higher than a certain level, a sound can be classified as ultrasound.

(2) It depends on the amplitude of sound waves.

(3) It is measured in decibels.

A (1) and (2) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

★3 7.1 A longitudinal wave travels from left to right in the air. The air particles vibrate and their positions at time $t = 0$ are shown in Figure b.

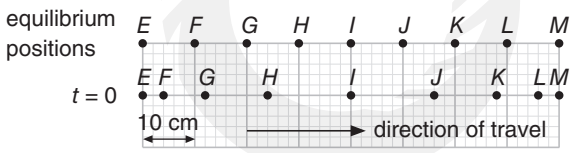


Fig b

Which of the following particles is moving to the right at $t = 0$?

A *F* B *H*

C *I* D *J*

★4 7.2 Which of the following phenomena serve as strong evidence that sound is a wave?

(1) Reflection of sound

(2) Refraction of sound

(3) Diffraction of sound

(4) Interference of sound

A (1) and (2) only B (1) and (3) only

C (2) and (3) only D (3) and (4) only

★5 Misc A wave travels from region *P* to region *Q*. Figure c shows the wavefronts of the wave.

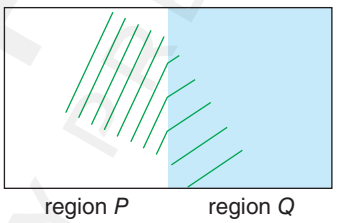


Fig c

Which of the following combinations is **not** possible?

	Type of wave	Region *P*	Region *Q*
A	EM waves	water	air
B	water waves	shallow water	deep water
C	sound waves	wood	air
D	light waves	glass	air

★6 7.1 A longitudinal wave is travelling from left to right in a medium. Figure d shows the positions of some particles (*P* to *V*) in the medium at different instants.

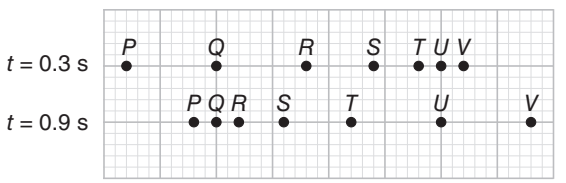


Fig d

Which of the following can be the period of the wave?

(1) 0.4 s

(2) 0.6 s

(3) 1.2 s

A (3) only B (1) and (2) only

C (1) and (3) only D (1), (2) and (3)

OXFORD UNIVERSITY PRESS

★7 7.2 In an experiment studying the interference of sound, two loudspeakers S_1 and S_2 emit sound waves in phase. Constructive interference occurs at P where S_1P = 2.4 m and S_2P = 4.4 m. Given that the speed of sound in air is 340 m s^{-1}, which of the following is **not** a possible frequency of the sound waves produced by the loudspeakers?

A 85 Hz
B 170 Hz
C 340 Hz
D 680 Hz

★8 7.2 A car travels at a constant velocity of 15 m s^{-1} towards a cliff (Fig e). The driver sounds the horn when the car is 500 m from the cliff and hears the echo after time t. Find t.

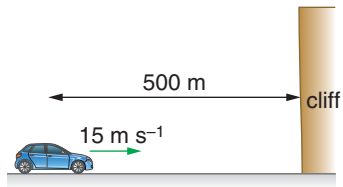


Fig e

A 1.41 s
B 1.47 s
C 2.82 s
D 2.94 s

★9 7.3 Figures f and g show the waveforms of two notes displayed on CRO screens. The CRO settings are the same in both cases.

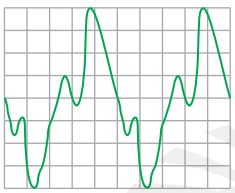


Fig f

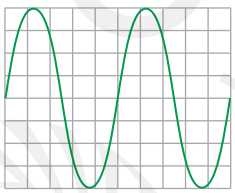


Fig g

Which of the following about the notes are correct?

(1) The two notes have different qualities of sound.
(2) The two notes have the same pitch.
(3) The two notes have the same loudness.

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

★★10 7.1 A longitudinal wave of period 2 s travels from left to right in a medium. Figure h shows the particles on the wave at time t = 0.

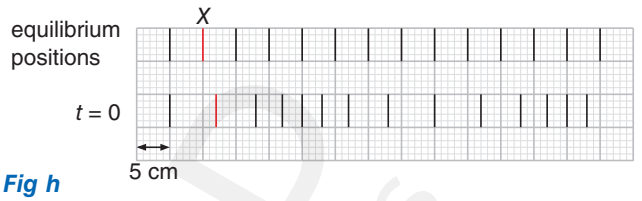
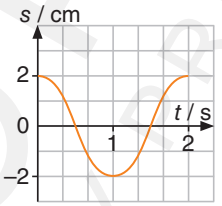
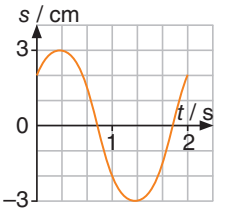
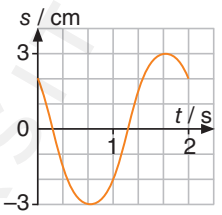


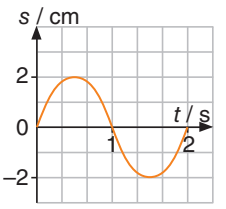
Fig h

Which of the following s–t graphs correctly describes the motion of particle X? Take the displacement to the right as positive.

A


B


C


D


Refer p.158, 161

★★11 7.1 A longitudinal wave is travelling from left to right in a medium. Figure i shows the positions of particles P to W in the medium at time t = 0.

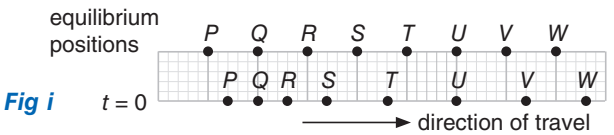


Fig i

Figure j shows the s–t graph of one of these particles.

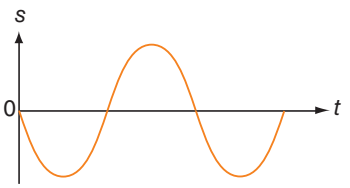


Fig j

Which particles may the graph describe?

A Q
B S
C U
D W

Refer p.158, 161

12 HKDSE 2014 Paper 1A Q17

Misc

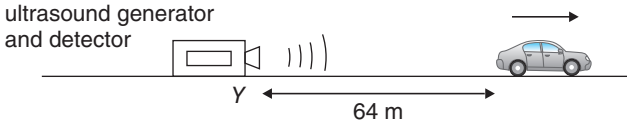


Fig k

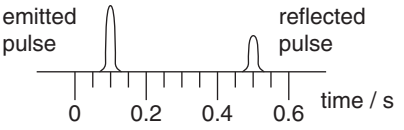


Fig l

Figure k shows a car travelling with a uniform speed along a straight road away from a stationary ultrasound generator and detector at Y. When the car is 64 m from Y, the generator emits an ultrasound pulse towards the car. The pulse is then reflected back to the detector at Y and displayed on a CRO as shown in Figure l. Estimate the speed of the car. Given: speed of ultrasound in air is 340 m s^{-1}

A 16 m s^{-1} B 20 m s^{-1} (53%)

C 24 m s^{-1} D 32 m s^{-1}

13 HKDSE 2016 Paper 1A Q21

7.2

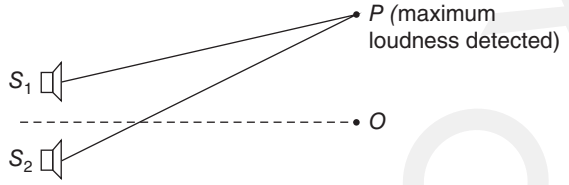


Fig m

Loudspeakers S_1 and S_2 connected to a signal generator emit sound waves which are in phase. Point O is equidistant from the loudspeakers while at point P maximum loudness is detected. The wavelength of the sound waves is λ. Which statement is **incorrect**?

A Both PS_1 and PS_2 must be integral multiples of wavelength λ. (45%)

B The definite value of the path difference $PS_2 - PS_1$ cannot be determined from the information given.

C At least one point of minimum loudness can be detected between O and P.

D Minimum loudness will be detected at P if the sound waves from S_1 and S_2 are in antiphase.

Q14 Exam report: 30% of the candidates wrongly thought that particle *I* was momentarily at rest at t = 1 s when it was at the equilibrium.

14 HKDSE 2017 Paper 1A Q14

7.1

Figure n shows the equilibrium positions of particles E to N in a medium. At time $t = 0$, a longitudinal wave starts travelling from left to right. At time $t = 1$ s, the positions of the particles are shown in Figure o.

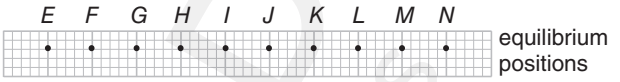


Fig n

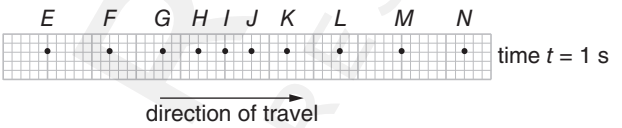


Fig o

Which of the following statements must be correct?

A The distance between particles F and N is equal to the wavelength of the wave. (54%)

B The period of the wave is 1 s.

C Particle E is always at rest.

D Particle I is momentarily at rest at $t = 1$ s.

15 HKDSE 2019 Paper 1A Q14

7.1

Figure p shows the displacement–distance graph at a certain instant of a longitudinal wave which travels to the right. Displacement to the right is taken to be positive.

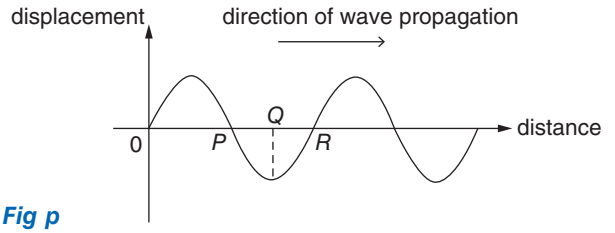


Fig p

At the instant shown, which of the following statements is/are correct?

(1) P is a centre of compression.

(2) A particle with its equilibrium position at Q is at rest.

(3) A particle with its equilibrium position at R is moving downwards.

A (1) only B (3) only

C (1) and (2) only D (2) and (3) only

(35%)

Q15 Exam report: Judging from the percentage of candidates choosing (3), nearly half of the candidates wrongly thought that the particle at *R* would move downwards for a longitudinal wave.

16 HKDSE 2019 Paper 1A Q17
7.2

Peter is standing next to the door of a room. He can hear the sound emitted by the television inside the room but cannot see the television pictures. Which of the following is/are the possible reason(s)?

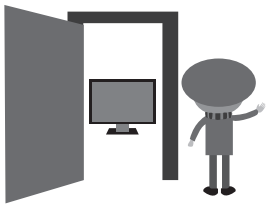


Fig q

(1) Sound wave diffracts while light wave does not.

(2) Sound wave is mechanical in nature while light wave is electromagnetic.

(3) Sound wave has a much longer wavelength than visible light.

A (1) only

B (3) only (50%)

C (1) and (2) only

D (2) and (3) only

Conventional questions

★**17** Misc

You are provided with two microphones connected to a timer, two tuning forks which produce sounds of different pitches and a rubber bung (Fig r). Describe how you would use the apparatus to conduct an experiment to show that the speed of sound in air is independent of frequency. (4 marks)

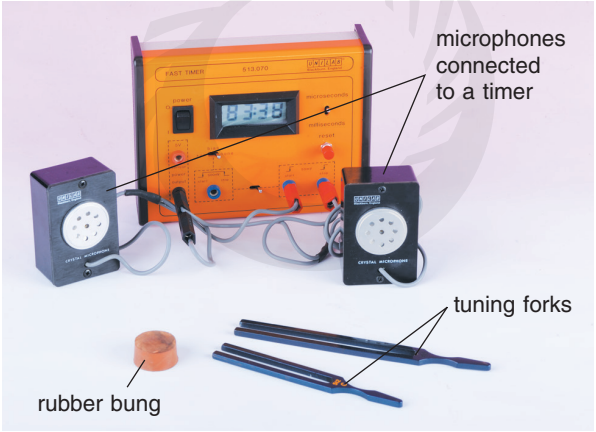


Fig r

★**18** Misc

Hummingbirds are small birds capable of hovering in mid-air by rapidly flapping their wings 12–80 times per second (Fig s). They are named after the characteristic humming sound created by their beating wings. The speed of sound in air is 340 m s^{-1}.

Fig s

(a) What is the frequency range of the humming sound produced by their wings? (1 mark)

(b) Can humans hear the whole frequency range of the humming sound in (a)? Explain briefly. (2 marks)

(c) Find the range of the wavelengths of the humming sound. (3 marks)

Shortest: 4.25 m; longest: 28.3 m

★**19** Misc

A girl hits the key of middle C (262 Hz) on the keyboard.

(a) Given that the speed of sound in air is 340 m s^{-1}, what is the wavelength of the sound produced? (2 marks)

(b) The girl then hits a key of a higher pitch. How would the wavelength of the sound differ from that in (a)? (1 mark)

(c) Figures t and u show the waveforms of two notes.

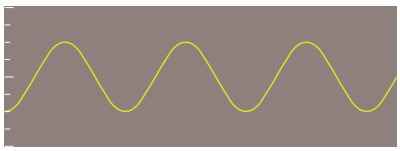


Fig t

Fig u

(i) Determine whether the two notes have the same pitch and loudness. Explain your answer. (3 marks)

(ii) Determine whether the two notes have the same quality. Explain. (2 marks)

★**20** 7.2 Philip sees a firework explode from 2 km away (Fig v). He hears the sound of the explosion 5.8 s after seeing the sparkle of the firework.

Fig v

(a) Explain the time delay between hearing the bang and seeing the flash. (1 mark)

(b) Estimate the speed of sound in the air using the data above. (2 marks)

(c) Suggest a reason why the firework operators at the firing spots should wear earmuffs. (1 mark)

(d) Mary cannot see the firework as her view is blocked by buildings. However, she can still hear the sound of the explosion. Explain. (2 marks)

★**21** 7.1 The *s*–*d* graph of a longitudinal wave travelling to the right at a certain instant is shown in Figure w, with the displacement to the right taken as positive.

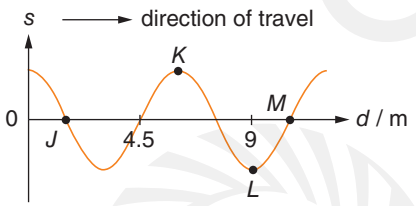


Fig w

Consider particles *J*, *K*, *L* and *M* on the wave at the instant shown.

(a) State the directions of motion of particles *K* and *M*. (2 marks)

(b) State the particle(s) at

(i) its equilibrium position;

(ii) the centre of compression;

(iii) the centre of rarefaction. (4 marks)

(c) The speed of the wave is 4500 m s^{-1}. Find the frequency of the wave. (2 marks)

★**22** Misc Soldiers in ancient times detected approaching enemies by putting their ears to the ground and listening (Fig x).

Fig x

(a) If a group of soldiers on horses were approaching from 10 km away, how much faster could a soldier using the above method detect their approach compared to a soldier standing up and listening through the air? Assume that sound travels through the air and the ground at 340 m s^{-1} and 4000 m s^{-1} respectively. (3 marks)

(b) Compare the frequencies and wavelengths of the sound waves travelling through the air and the ground. (2 marks)

★**23** 7.1 A sound wave is travelling towards the right in a material. The positions of the particles in the material at different instants are as shown (Fig y).

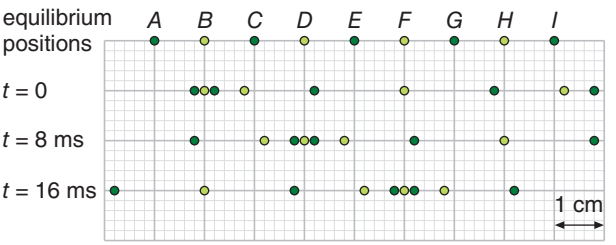


Fig y

(a) Determine whether the sound wave is within the audible frequency range. (3 marks)

(b) Draw the *s*–*d* graph of the wave at t = 8 ms. Take displacement to the right as positive. (2 marks)

(c) In which direction is particle *A* moving at t = 8 ms? (1 mark)

(d) Find the speed of the wave. (2 marks)

★24 Misc (a) State one similarity and one difference between sound waves and light waves. (2 marks)

(b) (i) A light ray travels from air to water (Fig z). Draw the refracted ray in the figure. (1 mark)

(ii) A series of sound waves travels from air to water (Fig aa). Draw the refracted waves in the figure. (2 marks)

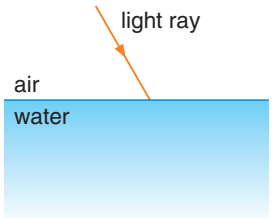


Fig z

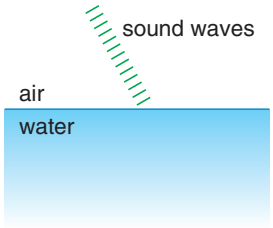


Fig aa

★25 Misc Two loudspeakers S_1 and S_2 are connected to a signal generator (Fig ab). The signal generator is set to 1.4 kHz. O is a point at equidistance from S_1 and S_2. As a student walks from O to Q, he hears a loud sound at O, then a soft sound at P and another loud sound at Q. The speed of sound in air is 340 m s^{-1}.

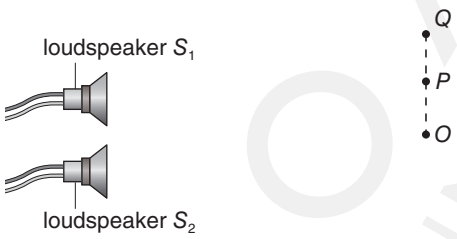


Fig ab

(a) Name the wave phenomenon that causes the change in loudness of the sound heard along OQ. (1 mark)

(b) Find the path difference at P and Q. (3 marks)

(c) One of the loudspeakers is removed and a gap is placed in front of the other loudspeaker (Fig ac). When the sound wave passes through the gap, it diffracts significantly.

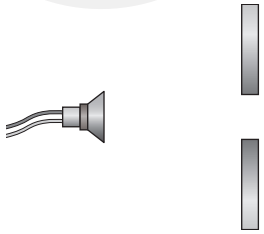


Fig ac

(i) What can you say about the size of the gap? (1 mark)

(ii) If sound with a higher pitch is produced by the loudspeaker, how would the degree of diffraction change? Explain briefly. (2 marks)

★26 Misc Julia sings a note and her voice is picked up by a microphone connected to a CRO. The CRO screen shows the waveform of her voice (Fig ad).

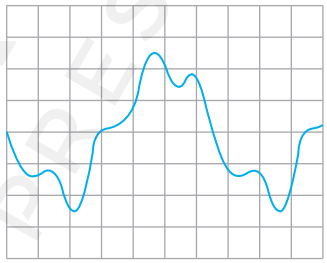


Fig ad

(a) Given that the time-base of the CRO is set to 5×10^{-4} s div^{-1} and the speed of sound in air is 340 m s^{-1}, estimate the wavelength of the note. (4 marks)

(b) Michael sings a note of the same pitch with a louder voice. Describe how his waveform on the CRO would be different from Julia's. (2 marks)

★27 Misc Some whales (Fig ae) emit ultrasound pulses to locate the position of their prey. The speed of sound in seawater is 1500 m s^{-1}.

Fig ae

(a) State the definition of ultrasound. (1 mark)

(b) How does a whale determine whether its prey is close by using ultrasound? (2 marks)

(c) If a whale emits ultrasound with a frequency of 120 000 Hz, find the wavelength of the ultrasound. (2 marks)

(d) Is it possible for ultrasound to undergo total internal reflection when it travels from the sea to the air? Explain briefly. (3 marks)

★28 Misc Read the following passage about **how we speak** and answer the questions that follow.

The vocal cords are two small elastic membranes located within our larynx (Fig af). When we breathe, the vocal cords form a 'V' shaped opening in the larynx, allowing air to go in and out of the lungs. When we speak, the vocal cords come together and the air expelled from the lungs passes through the larynx, causing the vocal cords to vibrate and produce sound.

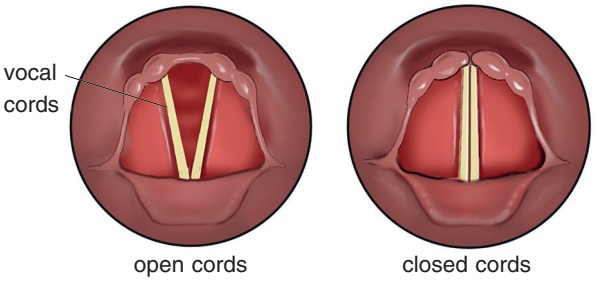


Fig af

We need to control the tension of our vocal cords to produce sounds at different pitches, and to control the amount of air through the larynx to produce sounds of different loudness. The greater the tension in the vocal cord, the higher the pitch of the sound; the larger the amount of air through the larynx, the louder the sound. In addition, we need to control the positions of our tongue, lips and teeth.

It has been found that the frequencies of sound produced by boys and girls range between 150–400 Hz. However, since the vocal cords of boys thicken during puberty, the range of the frequency of sound produced by mature men becomes 80–200 Hz.

(a) Explain briefly how sound is produced by the vocal cords. (3 marks)

(b) Describe briefly how a person can change the amplitude and frequency of vibration of the vocal cords. (2 marks)

(c) Find the longest wavelength of sound that can be spoken by a mature man.
Given: speed of sound in air is 340 m s^{-1}. (2 marks)

★29 Misc Two loudspeakers *X* and *Y* are connected to a signal generator set at 850 Hz (Fig ag). Point *O* is the mid-point of *XY*. At first, only loudspeaker *X* is turned on.

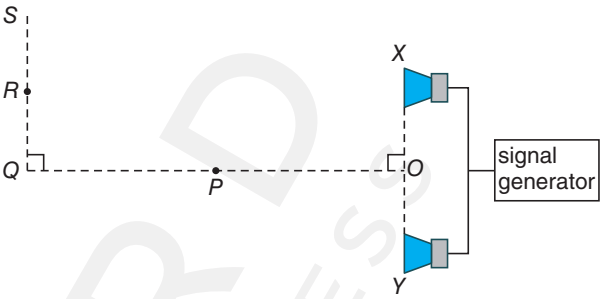


Fig ag

(a) The speed of sound in air is 340 m s^{-1}. Find the wavelength of the sound wave emitted by loudspeaker *X*. (2 marks)

(b) A sound intensity level meter is placed at *Q* to measure the loudness of sound. Sketch the variation of the meter reading as time passes. (1 mark)

Then both loudspeakers are turned on.

(c) The sound intensity level meter is moved from *Q* to *S*. Allan suggests that an ideal graph showing the variation of the meter reading is as shown in Figure ah.

Fig ah

(i) Explain why the loudness of the sound varies like this. (2 marks)

(ii) If background noise is taken into account, suggest a possible change to the graph. (1 mark)

(iii) In practice, maximum meter readings are detected only at points *Q*, *R* and *S*. The distance between *S* and *X* is 90 cm. Find the distance between *S* and *Y*. (2 marks)

(d) The sound intensity level meter is then moved from *P* to *Q*. Sketch the variation of the meter reading. (2 marks)

★★**30** 7.2 The speed of sound in air increases as air temperature increases. At a certain place, the air temperature varies with height. When Sam speaks from a high position, the sound travels to Tim on the ground along the path as shown in Figure ai.

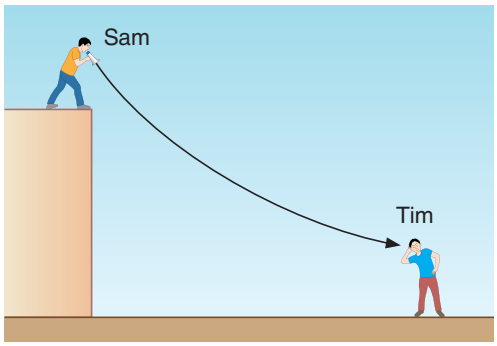


Fig ai

(a) Suggest how the air temperature varies with height at this place. Explain briefly. (3 marks)

(b) In Figure ai, mark the apparent position from where Tim would think the sound comes from. (1 mark)

(c) Now Tim speaks while Sam listens. In Figure ai, sketch the path of the sound travelling from Tim to Sam. (1 mark)

▶ Refer p.171

See p.188 for the Exam Report.

31 HKDSE 2018 Paper 1B Q7 Misc

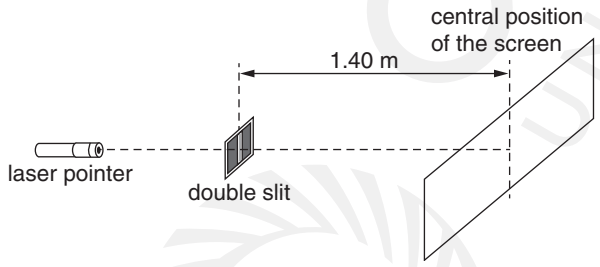


Fig aj

(a) Figure aj shows a set-up for measuring the wavelength λ of light emitted by a laser pointer. Several bright dots of average separation about 2 mm can be seen on the screen.

(i) For the same set of apparatus, suggest a way to increase the average separation between the bright dots on the screen. (1 mark)

The double slit is now replaced by a diffraction grating with 400 lines per mm.

(ii) Briefly explain why the accuracy of the experiment can be improved. (1 mark)

(iii) Only five bright dots are observed on the screen such that the separation between the 1st and 5th dots is 1.56 m. Find λ. (3 marks)

(b) To measure the speed of sound in air, a student connects two loudspeakers, A and B, to a signal generator as shown in Figure ak.

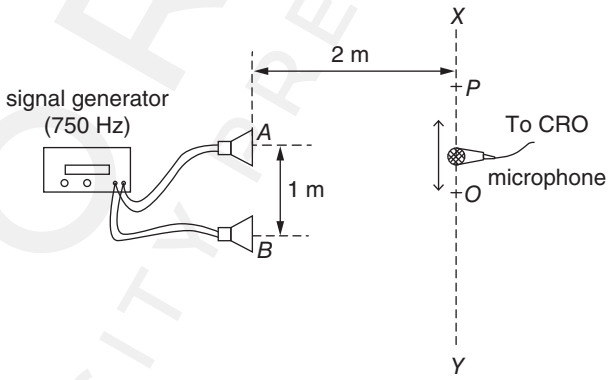


Fig ak

The separation of A and B is 1 m. A microphone is used to pick up the sound along the line XY at a distance of 2 m from the loudspeakers. The central maximum is at point O while the next maximum is at point P.

(i) With reference to the above settings, use the fringe separation equation $\Delta y = \frac{\lambda D}{a}$ in double-slit interference to find the wavelength λ of sound is not accurate. Explain briefly. (1 mark)

(ii) The distance between O and P is found to be 1 m when the signal generator is set at 750 Hz. By considering the path difference $PB - PA$, use the results of the experiment to find the speed of sound in air. (3 marks)

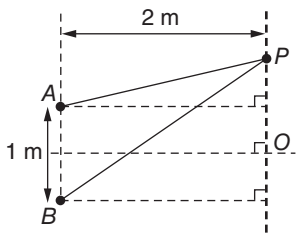


Fig al

See p.188 for the Exam Report.

32 HKDSE 2020 Paper 1B Q6

7.2

The set-up in Figure am is to find the speed of sound in air. Two identical microphones *A* and *B* are connected to a timer and placed on a bench top as shown. The timer can be triggered to 'start' and 'stop' timing using the respective microphones to feed signals to the START and STOP terminals of the timer.

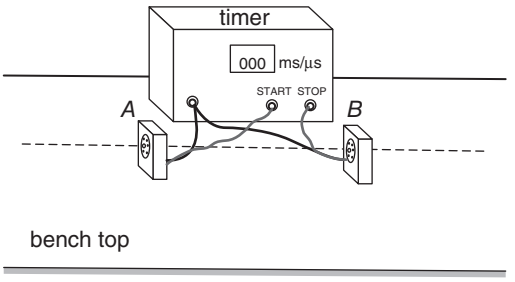


Fig am

(a) You are given a hammer and a metal plate ( ). Use 'X' to indicate a suitable location on Figure am where the hammer should hit the plate so as to generate a sharp loud sound to be received by the microphones in this experiment. Sate an additional piece of apparatus needed and the measurements to be made in this experiment. (3 marks)

(b) The separation between *A* and *B* is set at 0.280 m. The experiment is repeated to obtain a few readings of the timer as follows:

801 μs, 838 μs, 539 μs, 821 μs

(i) Find the speed of sound in air. Show how you would treat the data obtained in the calculation.

(ii) Suggest one adjustment to the experimental setting so as to obtain a more accurate result. (3 marks)

Know more

Loudspeaker (p.172)

In a loudspeaker unit containing more than one speaker cone, the larger speaker cone produces low-frequency sounds, while the smaller one produces high-frequency sounds (Fig a).

Sound waves of higher frequencies have shorter wavelengths. Using a smaller speaker cone increases the degree of diffraction and therefore allows high-frequency sounds to cover the same area as the low-frequency sounds.

Fig a

Noise-cancelling headphones (p.174)

Noise-cancelling headphones make use of the interference of sound to reduce noise. One or more microphones are placed inside the headphones to detect the noises from outside (Fig b). According to the signals from the microphones, the loudspeaker in the headphones produce sound waves that are in antiphase with the noises (in addition to the audio from the player connected to the headphones). Destructive interference occurs and therefore outside noises are reduced.

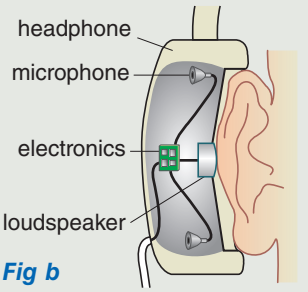


Fig b

→ More information can be found in the website.

Quality of sound (p.182)

A note is composed of a combination of frequencies (except the note produced by a tuning fork) (Fig c). The frequency and the relative amplitude of each component determine the waveform, hence the quality of the note.

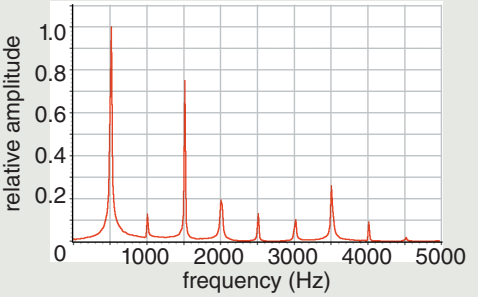


Fig c *Combination of frequencies produced by a flute.*